Financial Constraints and Trade Credit as a Strategic Tool: Evidence from Small-Scale Reservation Reforms in India

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Abstract

How do financial constraints affect an incumbent supplier firm's choice when facing an increased threat of entry from competitors? To defend market power, the threatened incumbent firms may (a) extend more trade credit, ex-ante, (b) reduce prices, or (c) do both. I test these predictions by exploiting plausibly exogenous, staggered removals of product–level entry barriers for Indian manufacturing firms. I find that the average incumbent firm strategically extends 11% more trade credit and lowers prices by 8.6%, to defend market power. Interestingly, firms with deeper pockets offer longer terms on trade credit, while financially constrained firms rely on price discounts.

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Introduction

Inter-firm credit, also known as *trade credit*, finances around 90% of the world merchandise trade, which amounts to $14 trillion (William, 2008). In the United States, for example, 80% of manufacturing firms offer their product on trade credit.[1] Recent empirical works have focused on understanding the role of trade credit supply in corporate bankruptcies as well as the probability of trade and the effect of trade credit supply on industry structure (Jacobson and von Schedvin, 2015; Breza and Liberman, 2017; Barrot, 2016).[2] However, the literature has given little attention to the reciprocal effect of industry structure on trade credit supply, especially its role in the context of defending product market power.

Whenever firms face an entry threat from new competitors, incumbent firms undertake various preemptive actions to deter entry.[3] Further, a large literature predicts that "deep-pocketed firms" attempt to drive financially constrained competitors out of business (Telser, 1966; Bolton and Scharfstein, 1990). One imperative question in this context concerns whether financial constraints affect firms' choice of strategic action (e.g., extending more trade credit versus offering price discounts) ex-ante when faced with a credible threat of increased competition.

The theory presents the following hypothesis here: an incumbent supplier firm may choose to (a) offer longer payment terms, (b) offer price discounts, or (c) offer both. But, the choice may vary with the financial strength of both the supplier and the customer firms. On the one hand, incumbent firms with deep pockets may choose to extend trade credit at longer payment terms to defend their market share when customers are financially constrained (i.e., when customers value a dollar of credit more than an actual dollar). These longer credit terms increase barriers to entry for potential competitors who have shallow pockets. Also, information asymmetry related to a customer's creditworthiness further dampens the potential entrant's ability to match the same strategy. On the other hand, the incumbent supplier firm may choose to offer

[1] See Tirole (2010).

[2] Jacobson and von Schedvin (2015), using Swedish data, find that the failures of trade debtor (customers) are associated with substantially enhanced bankruptcy risks for the trade creditors (suppliers). Breza and Liberman (2017), using large retailer data, find that the trade credit restriction reduces the likelihood of trade by 11%. Barrot (2016) finds, in the French trucking industry, that long payment terms impose a substantial *liquidity risk* on financially weaker suppliers and forces them into financial distress more often than if the weaker suppliers were paid earlier.

[3] Limit-pricing (Salop, 1979; Milgrom and Roberts, 1982), capacity building/investment (Dixit, 1980; Goolsbee and Syverson, 2008; Frésard and Valta, 2016), and leverage (Brander and Lewis, 1986; Cookson, 2017).

price discounts to its customers when the supplying firm is credit constrained and the customer firm is not.

It is challenging, however, to empirically test the above hypothesis in the data. The primary difficulty lies in cleanly identifying firms' choice of ex-ante strategic actions in response to a *threat* of entry as distinct from *actual* entry. First, theoretically, firms' trade credit policies, product pricing, and other product market strategies are jointly determined in equilibrium. Therefore, clean identification makes it essential to examine situations in which firms receive an *exogenous shock* in the product market. Second, ideally, this shock must affect the *threat of entry*, as opposed to the *actual entry* by a competing firm. This is an important distinction because *actual entry* by a competing firm is already an outcome of the trade credit strategies or price policies adopted by incumbent supplier firms. Moreover, the purpose of changing corporate policies such as trade credit policy or product prices after a rival has already entered could be very different from the incentives to change these policies before entry. For example, whether trade credit can act as a preemptive tool to deter entry cannot be examined if the rival firm has already entered or already produces the competing product (in the case of tariff cuts). In addition, a typical firm produces more than one product and has many inputs, so it is important to understand how a firm reacts when its main product line is under entry threat from competitor. Finally, when attempting to compare alternative strategies empirically, it is challenging to find joint data on trade credit, product prices, and sales and costs contributions at the product level.

I attempt to solve these identification and data challenges by exploiting the staggered removal of product-level entry barriers for Indian manufacturing firms as *plausibly* exogenous shocks to the threat of entry, which are unaffected by the existing policies of incumbent supplier firms. I find that, when the threat of entry increases, an average incumbent supplier firm extends 11% more trade credit and lowers prices by 8.6%. Also, I find that deep pocket firms offer more credit to their customers, and I find that financially constrained customers take more credit from their suppliers, consistent with the above hypothesis.

The institutional setting I examine concerns the removal of manufacturing restrictions for a list of products by the Government of India (GoI). These restrictions previously allowed only small-scale or large, export-oriented firms to manufacture products from this list.[4] Following

[4] The identification strategy is similar to Martin, Nataraj, and Harrison (2017), who find positive impact on

India's 1991 trade liberalization, the GoI appointed a special committee to reconsider the list of reserved items in 1995 (MSME, 2007). Based on recommendations from this committee, this reservation policy was dismantled in a *staggered* manner for about 1,024 products starting in 1997 (also known as *dereservation*). This resulted in the near-complete removal of reservations by 2008 (Figure 1).[5] For example, 15 products were dereserved in 1997, including ice-cream, poultry feed, hair dryers, ashtray.

The use of this particular institutional setting confers the following advantages. First, these product dereservations were arbitrary in nature. A reading of government documents, reports, and media does not give clear-cut reasons as to why certain products were initially reserved or dereserved at certain times. Even policy-makers in India now recognize the arbitrariness of this list of products subject to dereservation in various years (Hussain, 1997; Tewari and Wilde, 2017; Martin, Nataraj, and Harrison, 2017). In my analysis, I estimate a hazard model for the timing of dereservation, and I do not find evidence for existing trade credit policies of incumbent supplier firms predicting the event timing.[6] Second, the removal of entry barriers seems to have had real effects regarding increased competition which took effect slowly.[7] In a difference-in-difference setting, I find that the market concentration index (HHI) declined significantly within a few years of dereservation in the affected industries. This is consistent with the notion that it takes time for new entrants to install plants or expand their capacity to manufacture dereserved products. Consistent with economic theory, I also find a rising entry rate and a falling price index for dereserved products. This unique setting allows me to distinguish firms' ex ante strategies when facing the threat of entry. I specifically focus on a short window around dereservation to identify firms' preemptive actions when facing an increased *threat of entry.*

Using this unique regulation change that affected Indian manufacturing firms, I implement

employment growth after the removal of entry barrier.

[5] Internet Appendix (IA), Table IA1 shows sample notification about list of items dereserved in 1997, while Table A2 provides information on timing of dereservation by industry-year. Note that by end of 2008 only 20 products were in the reserved category and in April 2015, the last 20 products were removed from the reservation list.

[6] In a robustness check, in treatment group, I keep products that are not produced by large leading firms in the country and find similar results.

[7] Survey of small firms by Federation of Indian Chambers of Commerce and Industry (FICCI) indicates that small-scale firms see entry of multinational corporations (MNCs) and dereservation of the items reserved for them as a threat to their market share (http://www.business-standard.com/article/specials/ssis-feel-threatened-by-mncs-dereservation-ficci-survey-197122901002_1.html).

a difference-in-differences approach to estimate the effect of dereservation on the provision of trade credit and product prices. Specifically, I examine whether *incumbent* firms (i.e., firms that produced the reserved product before the reform) extend more credit ex-ante to their customers when they face an increased entry threat from new competitors (i.e., firms that might have moved into the product space after that product was dereserved). I find that the average incumbent supplier firm extends around 11% more trade credit (measured as the ratio of accounts receivable to sales) surrounding the removal of entry barriers compared to their counterparts, who do not face any such change in entry barriers. For the average incumbent supplier firm with trade credit payment terms of about two months, this translates to an increase in payment terms of around 8-10 days. Also, in the pre-dereservation period, I find no differences in the ratio of accounts receivable to sales (a proxy for trade credit terms) between (a) firms whose products were dereserved and (b) a group of control firms whose products were neither reserved nor dereserved. In addition, I find that for the average firm, the price of the firm's dereserved primary product also decreased by 8.6%. The incumbent supplier firms choose to extend trade credit at longer terms and also cut prices in anticipation of increased competition. This finding is robust to the inclusion of industry-year fixed effects, different sample periods, placebo samples, and alternative definitions of treatment and control firms.

I then test how the choice of extending trade credit versus price discounts as a preemptive strategic tool varies with the financial strength of the supplier firm. Extensive literature predicts that deep-pocketed firms attempt to drive financially constrained competitors out of business (Bolton and Scharfstein, 1990). In particular, long payment terms impose substantial *liquidity risk* on financially weaker suppliers, and these longer terms force weaker suppliers into financial distress more often than they would if they were paid earlier (Barrot, 2016). Consistent with this view, I find that large and old firms (using the size-age index proposed by Hadlock and Pierce (2010)) offer longer terms of credit, while financially constrained firms opt for price discounts. I find that firms with above-median size-age index, increase their trade credit terms by 22 days, which is equivalent to reducing prices by 6.92%.[8] On the other hand, firms with

[8]For above-median size-age index firms one year after the dereservation trade credit granted to sales increase by 0.061, which is equivalent to 22 days (0.061×365). A survey conducted for 141 Indian firms by the Center of Analytical Finance, Indian School of Business, finds that a typical discount for payment within 15 days is about 4% (De and Singh, 2013). The cost of trade credit for a customer firm with 73 days (average term before dereservation) is about 25.17% ($0.04\times365\div(73-15)$). With a 22-day increase in the term of credit, one year after the dereservation, the cost of trade credit for a customer firm is about 18.25% ($0.04\times365\div(95-15)$). This

less access to credit (i.e., firms with below-median size-age index) cut prices by 28%.

In addition to trade credit, the supply of credit from banks is an important source of financing for Indian firms. For the median firm, 25% of its assets are financed through bank credit. I find that firms with more banking relationships at the time of entry threat offered more credit to customers, while incumbent firms with fewer banking relationships rely on price discounts. Interestingly, I find that only large-old firms and those with many banking relationships receive more bank credit after dereservation. These results further validate the use of size-age index or prior number of banking relationships as a proxy for access to financing. I also find consistent results using various other proxies for access to finance like relationship with large banks, access to bank credit measured using firm geographic location, and supply of credit from product suppliers in the form of trade credit. To further confirm these results, I use a quasi-natural experiment: The establishment of debt recovery tribunals (DRTs) in India (Gopalan, Mukherjee, and Singh, 2016), that reduced the cost of the legal enforcement of debt contracts in different states without any accompanying change in the underlying law. I find similar results using this as a proxy for access to financing.

It is also important to understand why customer firms accept trade credit rather than price discounts. Although I do not observe the customers of the sample firms, I can observe the sample firms as customer firms and test whether they receive more credit when their suppliers face competition. In other words, I consider the trade credit received by sample firms after the dereservation of products used as their main input (i.e. their raw material). I find that when sample firms' primary input is dereserved, they receive more trade credit and price discounts. Interestingly, I find that only financially constrained firms (identified using their size-age index and establishment of DRTs in the state) accept more trade credit, while unconstrained firms accept price discounts.

In India, the reservation of products to be manufactured by small firms is not the only policy used in the country to protect small firms. In addition to protection from large manufacturing units, small firms in the country receive priority for bank credit. All banks (public and private) are required to lend at least 40% of their net credit to the "priority sector," which includes agriculture, agricultural processing, transport, and small-scale industry (SSI). If banks do not satisfy this priority sector target, they are required to lend money to specific government

is equivalent to a price discount of 6.92%.

agencies at very low rates of interest. If they lend to the priority sector, they lend at the same interest rate as in other sectors (Banerjee and Duflo, 2014).

During my sample period, the definition of small firms for priority sector lending changed three times. I use this heterogenity to identify how incumbent firms in the priority sector react to the threat of entry. The evidence suggests that small firms within the priority sector not only cut prices but also offer trade credit to defend their market share.

Overall, my analysis contributes to a few strands of literature. First, my analysis closely relates to Barrot (2016), who uses exogenous variation in the trade credit supply to identify the causal effect of trade credit supply on industry dynamics. In contrast, I examine the other side of this interaction: I estimate the causal effect of entry threats on firms' choice of extending trade credit. In this sense, our papers complement each other. Also, I provide comparisons between the choice of trade credit versus price discounts when a supplier/customer firm faces financial constraints, thereby enriching our understanding of how and when firms use trade credit as a strategic tool. Further, the joint availability of data on trade credit, product prices and sales and costs contributions at the product-level further enhances the contribution of this study to the existing literature.

Second, my study also relates to a growing literature that documents how incumbent firms respond to the threat of entry. The theoretical literature in this domain discusses various strategic tools and rationales for *preemptive* actions (Dixit, 1980). On the empirical side, Frésard (2010) documents the impact of cash on product market performance; Hoberg, Phillips, and Prabhala (2014) examine how product market threats influence firm payout policy and cash holdings; and Frésard and Valta (2016) find that investments by domestic US firms decrease in response to the lower cost of entry for foreign firms. Parise (2017) and Goolsbee and Syverson (2008) document how incumbent carriers adjust debt maturity and prices, respectively, when the probability of future entry by low-cost air carriers in their route network dramatically increases. Cookson (2017) finds physical capacity expansion by low-leverage incumbent US casino firms when they face an entry threat. In this context, my study examines the role played by financial constraints for strategic adjustments in trade credit versus price.

Third, this paper relates to studies that examine the role of product market competition in corporate finance. In particular, it complements the few empirical papers that relate trade credit to product market structures (Fisman and Raturi, 2004; Hyndman and Serio, 2010;

Barrot, 2016).

My study incrementally and substantially contributes to the abpve literature along four distinct dimensions. First, I rely on plausibly exogenous variations from the removal of entry barriers at the product level to cleanly identify the causal effect of product-market entry threats on firms' supply of trade credit and product prices. Second, I concentrate on firms' trade credit response to entry threats, as opposed to credit supply behavior following entry. This distinction is important because it allows me to examine the value of trade credit as a preemptive strategic tool without influence from other confounding factors associated with actual changes in competition (e.g., changes in a firm's financial position). Third, I can estimate the firm's exact *exposure* to the entry threat because I can observe the firm's sales revenues across various products sold and quantities produced. The availability of input-output data and prices further enhances the richness. In addition, the product-level sales data allow me to control for time-varying unobservable characteristics at a greater granularity, thereby aiding identification.[9] Finally, the institutional setting allows me to examine *plausibly* exogenous shocks to the threat of entry across a broad spectrum of industries rather than focusing on a single industry.

The remainder of this paper is organized as follows. Section 1 reviews the theory and evidence on trade credit and provide details on the main hypotheses. Section 2 presents the dereservation reforms, which serve as my main source of identification. Section 3 presents the data and methodology. Section 4 describes the results. Section 5 describes the robustness of results. Section 6 concludes.

1 Trade Credit Theories and Hypotheses

In this section, I survey various trade credit theories and how they are related to product market competition. Then, I use these theories to develop my hypotheses.

1.1 Trade Credit Theories

The trade credit literature has expanded in two directions: a financial motive approach and a product market competition approach.

[9] I define a firm's product using the 5-digit National Industrial Classification (NIC) code. Therefore, I can control for time-varying unobservable characteristics in the 4-digit code. There are 2,710 products, defined using the 5-digit NIC codes, linked to 137 4-digit NIC industries across the 24 manufacturing sectors (2-digit NIC codes). In comparison, US data used by Bernard, Redding, and Schott (2011) contain approximately 1,500 products, defined as 5-digit Standard Industrial Classification (SIC) codes, across 455 4-digit SIC industries.

The *financial motive* approach to trade credit explains how credit rationing affects the demand for trade credit.[10] This approach has explored the motives of the "lender" and "borrower" of trade credit compared to other financial sources. It has focused on the problem of why a lender gives trade credit to a buyer, and why a buyer chooses to "borrow" trade credit instead of some other type of borrowing. This approach implicitly assumes that financial motivation leads to trade credit provision.

Petersen and Rajan (1997) documents a comprehensive survey of theories and empirical tests on *financial motive* for trade credit. They use the SME data in the United States, and find that suppliers are inclined to lend to financially constrained customers. One interpretation of this result is that suppliers offer goods on credit because they have more information about their buyers. Burkart and Ellingsen (2004) suggest that suppliers have an informational advantage that banks do not have. They hypothesize that it is typically less profitable for an opportunistic borrower to divert inputs than to divert cash, and this tendency increases the advantage of suppliers over banks in lending to their clients. This advantage that suppliers have over external financiers may also be nested in the nature of the underlying product. Fabbri and Menichini (2010) theorize that the liquidation advantages of the supplier over other financing sources allow them to provide credit. This comparative advantage is more pronounced for differentiated goods, because these are often tailored to the needs of a small group of customers.

The *product market competition* approach concentrates on how competition in the product market affects trade credit provision. The literature presents contrasting evidence in this context. On the one hand, McMillan and Woodruff (1999) find that the presence and number of competitors within a 1-kilometer radius reduce trade credit provisions to customers. Similarly, Johnson, McMillan, and Woodruff (2002) find that trade credit provision is lower when more than 5 rivals are less than one kilometer away. Their results suggest that competition prevents suppliers from providing credit to avoid risk. On the other hand, subsequent studies find an opposite result. Fisman and Raturi (2004) document that the monopoly power of the supplier is negatively associated with credit provision, which counters the assertions of previous studies that monopoly power facilitates the provision of credit because monopolists are better able

[10] Some other theories of trade credit suggest that trade credit reduces transaction costs (Ferris, 1981), allows price discrimination between customers with different credit worthiness (Brennan, Maksimovics, and Zechner, 1988), provides a warranty for quality when customers cannot observe product characteristics (Long, Malitz, and Ravid, 1993), and fosters long-term relations with the customer (Wilson and Summers, 2002).

to enforce payment. Later, Hyndman and Serio (2010) show that the relationship between trade credit provision and supplier's market power is not linear but rather follows an inverted U-shape. A monopolist supplier often prefers to sell only in cash, which is zero trade credit. Once competition starts, trade credit grows with the number of competitors. Hyndman and Serio (2010) argue that this occurs as Bertrand price competition in the cash market raises the price of cash, thus new entrants can only offer trade credit due to the product market competition. With the intensification of competition, problems of commitment on trade credit repayment and decisions on credit provision become less important. However, enforcement becomes constrained as the number of competitors increases and outgrows a certain limit.

In this paper, I combine both motives of trade credit. I show how a firm's choice of trade credit versus price reduction as a strategic tool varies with financial constraints. The above *product market competition* theories all propose arguments on how industry structure affects trade credit. Barrot (2016) documents the reverse: how trade credit provision affects industry dynamics. He finds that restrictions on trade credit provision affects the entry/exit rates of trucking firms in France. He concludes that longer terms extended by financially stronger firms may thus serve as a barrier to entry and prevent their constrained,–yet potentially efficient– rivals, from entering, expanding, and surviving in the industry. However, no one has explored whether incumbent firms use trade credit, price discounts or both, ex ante, when they an face entry threat from competitors.

1.2 Hypotheses Development

The literature above suggests that incumbent supplier firms should extend more trade credit before the potential entrant starts manufacturing the product. In other words, incumbent supplier firms should lengthen the payment terms of trade credit to their customers when they face an entry threat due to the removal of entry barriers. Trade credit raises barriers for the potential entrant. However, if incumbents can easily adjust trade credit terms after the entry, then longer payment terms to customers may not serve as a credible strategic tool for entry deterrence. Barrot (2016) finds long payment terms (measured as the ratio of accounts receivable to sales) impose a substantial liquidity risk on financially weaker suppliers, and force them into financial distress more often than they would if they were paid earlier. This suggests that an increase in trade credit may not be easy to reverse. Therefore, extending trade credit is an effective signal that is both costly and credible. This suggests that an effective increase in competition encourages incumbent firms to increase trade credit. At the same time, the limit-pricing theories (Salop, 1979; Milgrom and Roberts, 1982) suggest that similar benefits can be given to customers by offering a lower price, and this can also help deter entry. This suggests that incumbent firms may also use price discounts as a strategic tool.

H1 Price vs. Trade Credit Hypothesis: With an increased threat of entry, incumbent firms prevent their customers from switching to their new suppliers by (a) increasing trade credit, (b) reducing prices, or (c) both.

A natural extension relates to the incumbent supplier firms' ability to finance trade credit. A large literature shows that deep-pocketed firms attempt to drive financially constrained competitors out of business (e.g., Telser (1966); Bolton and Scharfstein (1990)). Therefore, when incumbent supplier firms expect greater competition, ex-ante financially strong incumbent firms may extend more trade credit, while financially constrained firms may use limit-pricing as a strategic tool.

H2: Supplier's Financial Strength Hypothesis: With increased threat of entry, financially strong (weak) incumbent supplier firms *increase trade credit* (*decrease prices*) to influence the entry decision of the potential entrants and prevent customers from switching to new suppliers.

At the same time, when a supplier offers more credit, customer firms will accept it only if they are financially constrained, otherwise they choose the price discount. Financially constrained customer firms, who value a dollar of credit more than an actual dollar, take more credit from their supplier, when supplier firms face greater product market competition.

H3: Customer's Financial Strength Hypothesis: With an increased threat of entry for supplier firms, financially weak (strong) customers *accept more trade credit* (*ask for price discounts*) from supplier firms.

2 Dereservation Reform

The Indian government has a long history of promoting small–scale industries. Starting in 1960, the Government of India reserved a large number of manufactured goods for exclusive production by small scale firms. The production of hundreds of products across the manufacturing sector were restricted to small–scale firms, insulating them from competition.[11]

It was argued that small establishments producing labor–intensive goods would absorb the abundant labor supply present in an underdeveloped country. However, in official documents, there is no clear criterion for the selection of goods to be reserved. In addition, there is no mention of other criteria for reserving specific goods, for example, based on optimal capital–to–labor ratios (which are difficult to ascertain in the first place). For example, in the garment industry, cotton and woolen socks, scarves, clothes and vests were reserved, while linen, jute or hemp products were not reserved. This suggests a high degree of substitutability between reserved and non-reserved clothing items. The types of products on the reserved list were varied, spanning many industrial sectors such as food, chemicals, electronics, and textiles. Within the small–scale sector, the output share of reserved products was approximately 30% in 1987. Overall, reserved products constituted about 12% of Indian manufacturing output. There was considerable heterogeneity across industry sectors, with reserved products forming 80% of output in hosiery and garments, 57% in certain wood products, and a negligible fraction in textiles.[12]

Despite India's liberalization of a variety of industrial and trade policies in 1991, the reservation of products for small–scale sector remained in force until the late 1990s. As of July 1991, there were around 800 items, which constituted around 1,024 products in the *reserved* category.[13] According to this policy, only small–scale firms could manufacture these 1024 products. The large– or medium–sector firms that already made the reserved products were permitted to continue to manufacture within the reserved product category, but their output was capped at current levels. They could expand their capacity only if they agreed to export a minimum of 75% of their production from fresh capacity. This regulation induced heterogeneity in the reserved sector, since it included both small firms and large, export-oriented firms.

The Advisory Committee on Reservation recognized growing concerns about small–scale

[11] The Indian government defines a *small–scale firm* according to the cumulative amount of investment in plant and machinery. This means that all the plants with a level of capital below a limit set by the government are considered "small" and are therefore allowed to produce reserved goods. These limits have changed over time. It started at INR 500,000 in 1960 and has been periodically adjusted upward using inflation (García-Santana and Pijoan-Mas, 2014).

[12] An expert committee, the Abid Hussain Committee constituted in 1997 to review small–scale industry in India's post–liberalization period, states, "The choice of products for reservation was necessarily arbitrary"(Hussain, 1997).

[13] According to Notification S.O 477(E) issued by the Ministry of Micro, Small and Medium enterprises (MSME), Govt. of India. http://www.dcmsme.gov.in/publications/circulars/477E.pdf

sector policies that followed the 1991 trade liberalization. The small–scale firms had to compete with imported goods, and large firms might be able to exercise monopoly power in the market for reserved goods, as most other producers would be small. Moreover, growing consumer demand for high-quality goods and ongoing technological progress made it more difficult to produce many items in this sector. The Advisory Committee therefore appointed a special committee to reconsider the list of reserved items in 1995 (MSME, 2007). Based on recommendations from this committee, most of the 1,024 products were dereserved starting in 1997 (Figure 1).[14]

The identification of the threat of entry in this paper comes from a rapid and complete dismantling of the reservation policy. Large–scale dereservation started slowly in 1997 (15 products) and accelerated in 2002 (51 products). From 2003 to 2008, approximately 100-250 products were dereserved each year. Finally, in April 2015, the last 20 products were removed from the reservation list. Internet Appendix Table IA1 shows a sample notification about the list of items dereserved in 1997.[15] For example, 15 products were dereserved in 1997, which included ice–cream, poultry feed, hair dryers, and ash tray.

As evident from Internet Appendix Table IA2, the dereservation of products is staggered across time and industries. To give an overview of the variation across industries, from 2003 to 2008, a total of 809 products in over 24 (2-digit NIC) manufacturing industries were dereserved, and among these, chemical products contributed the most (30%). For example, products in Basic Metals (2–digit NIC Code = 24) were never dereserved during this period. However, 107 products (13.23% of total 809 products) in Fabricated Metals (2 digit NIC Code = 25) were dereserved. Similarly, 13 wooden products (1.61% of the total, 2-digit NIC Code = 16) were dereserved, while 11 furniture products (1.36% of total, 2-digit NIC Code = 31) were dereserved.

A similar staggered variation in dereservation is also evident across industries over time. As discussed before, products in the Basic Metals industry were never reserved/dereserved, while products in the Fabricated Metals industry were dereserved in a staggered manner as follows: 0% (2003), 22% (2004), 0% (2005), 42% (2006), 31% (2007), and 5% (2008). A similar staggered variation is evident in other industries over time, and Internet Appendix Table IA2)

[14] For a more detailed description of the process of dereservation, see http://dcmsme.gov.in/publications/reserveditems/itemrese.htm.

[15] All the notifications about dereservation are available at http://www.dcmsme.gov.in/publications/notiissforresderes.html

gives more information about the timing of dereservation across industries over time.

Was dereservation really random? One could be concerned about the potential anticipation of the dereservation policy. Anecdotal evidence and discussions suggest that the reservation of products and the timing of their dereservation appear to be arbitrary in nature and varied across different industries. As in the case of reservation, the process of dereservation is also not well understood. A reading of government documents, reports, and media does not give clear-cut reasons as to why certain products were initially reserved or dereserved at certain times. Martin, Nataraj, and Harrison (2017) and Tewari and Wilde (2017) document that a conclusive and definitive account of the dereservation process is not available. Explanations range from competition from imports, technology requirements, the need to comply with regulations, a lack of any benefit to small producers, or the availability of unreserved substitute products. A product's path to dereservation tends to be lengthy and circuitous. A product is identified as a dereservation candidate by a ministry or by industry players (including manufacturers of reserved products themselves who find the investment ceiling constraining). Once identified, a series of meetings are held between "stakeholders" (such as trade associations or small firm groups and officials). After a review from a chain of bureaucrats, the dereservation of a product is signed into law by the central government minister. Qualitative support for the above arguments on the nature of reservation and dereservation is reflected in the extent of reservation/dereservation both across and within product categories.

One example of the arbitrary nature of reservation/dereservation can be found in the reservation of various oils. Many oils were never reserved, while several oils were reserved, such as hair oil, sesame, and mustard oil. However, they were dereserved at different point in times such as hair oil was dereserved in 2003, sesame oil in 2008, and mustard oil in 2015. Similarly, among leather products, leather shoes were dereserved in 2001 while leather slippers were dereserved in 2003. It is difficult to think of some systematic and endogenous rationalization in this cross-section and time-wise pattern of reservations.

3 Data and Methodology

3.1 Firm Characteristics

I compile a firm-level panel data set spanning 1990 to 2013, from the Prowess$_{dx}$ database collected by the Centre for Monitoring the Indian Economy (CMIE).[16] This database contains information primarily from the income statements and balance sheets of about 34,197 publicly listed and private companies, of which 10,724 are in the manufacturing sector. This database is a firm-level panel that also records detailed annual information on firms' product mix.[17] Furthermore, for each product manufactured by the firm, the data set provides the value of sales, quantity, and units. With this data, one can easily create a price index for each product for a given firm-year. In addition to output, the data set provides information on all the materials used to manufacture the reported products. The Prowess$_{dx}$ is therefore particularly well suited for understanding how firms adjust their financial policies and price policies when their product lines are under threat of competition. Also, with the availability of the input data, one can look at a firm's behavior as a customer. The definition of a product is based on the CMIE's internal product classification. I complement the data on firm product mix with measures on product reservation policy at the 5-digit NIC (similar to SIC of US) product level. I follow Goldberg, Khandelwal, Pavcnik, and Topalova (2010) to clean up the product-level database. Internet Appendix IA1 provides more details on product data and sample selection. I follow the sample firms from beginning of 1990 to the end of 2013. All financial variables are inflation adjusted using the Wholesale Price Index (WPI) at 2004-2005 prices. I also correct for changes in the financial reporting year. In my robustness checks, I start sample from the beginning of 1997 to ensure that my estimates are not influenced by industry-level trade liberalization policies, since these changes were completed before 1997. Thus, my estimates are robust to sample selection period.[18] Since the reservation policy was intended only for manufacturing

[16] Prowess$_{dx}$(http://prowessdx.cmie.com), is special data extraction interface for academicians, similar to Prowess. This data set has been used by a number of prior studies on Indian firms, including Bertrand, Mehta, and Mullainathan (2002); Gopalan, Nanda, and Seru (2007); Lilienfeld-Toal, Mookherjee, and Visaria (2012); and Gopalan, Mukherjee, and Singh (2016).

[17] Indian firms are required by the 1956 Companies Act to disclose product-level information on capacities, production, and sales in their annual reports. The CMIE compiles these detailed quantitative data.

[18] Major changes in policies vis-à-vis foreign investment occurred in the early 1990s and then stalled during the period of dereservation reform. Nataraj (2011) shows that tariffs were largely harmonized across industries by the late 1990s, so even though there were some reductions during the 2000s, the variation in tariff rates

firms in India, I keep only non-service, and non-government firms in my sample. Appendix A provides the definition of all the variables. Internet Appendix IA1 and IA2 further details on sample selection, matching product data with financial data and variable construction. Table IA3 and IA4 details on matching of product data.[19]

3.2 Measure of Trade Credit Supply

Following Petersen and Rajan (1997), I proxy payment terms by the amount of accounts receivable on firms' balance sheets as recorded at the end of their year of operations *scaled by* total cumulative sales in the year (Accounts Receivables/Sales). A higher ratio implies that a significant amount of cash is tied up. In other words, an increase in accounts receivable to sales ratio from one year to the next indicates that investment in the accounts receivable is growing more rapidly than sales.

This widely used measure of trade credit provision is a rough proxy for actual payment terms. A seasonal business experiences a large part of its annual sales in a particular part of the year, so this ratio can be seasonal. However, it is unlikely that this seasonality in the sample changes dramatically following the dereservation reform. In addition, the inclusion of industry-year fixed effects at a more granular level helps absorb seasonality within industries across time. Similarly, this ratio may mechanically overestimate the average payment terms in periods of growth and underestimate them during downturns. This makes it important to have an appropriate control group. The staggered nature of dereservation reform helps me use the pre-reform period observations of treatment firms as an appropriate control group, and this aids identification of the effect of entry threat on payment terms.

In addition to Accounts Receivables/Sales, I estimate my results using Log (1+ Accounts Receivables) to further confirm the findings. To ensure that my results are not affected by outliers, I winsorise all the ratios at 1% and omit observations with a ratio of accounts receivables

across product types had fallen dramatically by the start of 1997.

[19]Unlike Martin, Nataraj, and Harrison (2017); who use ASI data, I rely on the Prowess database due to the joint availability of information on trade credit, price data, and input and output data for a given firm. In addition, ASI provides establishment–level information, while in this study, I am more interested in firm-level strategic actions. I understand that I may not be capturing very small and micro enterprises that were manufacturing the reserved products, but the data capture relatively medium– and large– sized incumbent firms that were capacity constrained. And with the removal of entry barriers, it remains interesting to see how capacity–constrained incumbents respond. The Prowess data do not provide much information on employment and wages, which was the interest of the Martin, Nataraj, and Harrison (2017) study.

to sales greater than 1.

3.3 Summary Statistics

As reported in Table 1, the median Accounts Receivables/Sales ratio is about 0.16 with a mean of 0.19, which is equivalent to the average payment period of about 70 days (=0.19×365 days). The median firm's price of output measured as *Log (Primary Product Price Index),* is 0.46, which translates to a price index of 1.25. For the average firm, 18% of assets are financed using trade credit. In addition to trade credit, the supply of credit from banks is an important source of finance for Indian firms. For the median firm, 25% of assets are financed through bank credit. The median firm size, measured as *Total Assets* in my sample is $11.17 Mn., which translates into a book value of total assets amounting to INR 503 million at 2004-2005 prices (with $1=INR 45, exchange rate). The 10th percentile and 25th percentile firms have book value of total assets amounting to INR 77 million and INR 180 million, respectively, while the minimum is about INR 0.88 million. Firms in India have little cash on their balance sheet, as seen by the mean value of cash holdings of 6%. Firms in my sample are profitable, as seen from the mean value of profitability (EBIT/Assets) of 12%. On average, sales growth is about 14%.

In my sample, a firm produces a median of two products, with a mean of 2.7 products. The inter-quartile range for sales revenue from the primary product varies from 70–99%.

3.4 Methodology

To identify the incumbent supplier firms that are affected by the threat of entry, I define treatment as the elimination of small-scale reservation on the incumbent firm's primary product in the NIC 5-digit category. A firm's primary product is the product that has the maximum sales revenue in the year that immediately precedes dereservation. I start with a difference-in-differences equation of the following form for firm i in year t:

$$\mathrm{y}_{it} = \alpha_0 + \sum_{k=-2}^{-10} \beta_k Pre-Dereservation(k)_{it} + \sum_{k=0}^{10} \beta_k Dereservation(k)_{it} + \alpha_i + \alpha_t + \varepsilon_{it}. \quad (1)$$

The dependent variable y_{it} is defined as the *accounts receivables scaled by sales* of firm i at time t. The variable *Pre-Dereservation(k)*$_{it}$ (*Dereservation(k)*$_{it}$) is a dummy that takes a value one if it is k years before (after) the incumbent firm's primary reserved product has been dereserved. The *Dereservation(k)*$_{it}$ dummy identifies the group of firms under threat of entry. In other specifications, I replace the dummy with the exposure of the incumbent firm to the entry threat by using the proportion of total sales revenue from dereserved products in pre-dereservation period.

An incumbent firm is defined as a firm that has manufactured a reserved product before it was dereserved.[20] In certain specifications, I include all firms (even those that do not help to identify β_k because they are not affected by the reservation policy) because these firms help to identify the secular year trends in firms' financial policies.

Also, *Pre-Dereservation (-10)* equals one if it is 10 or more years before dereservation of the product, and *Dereservation (+10)* equals one if it is 10 or more years after dereservation.[21] The model is fully saturated with the year immediately before the dereservation as the excluded category. Therefore, the coefficients on *Pre-Dereservation(-k)* (*Dereservation(k)*) compare the level of the dependent variable 'k' years before (after) the dereservation to the year immediately before its dereservation.

The inclusion of firm fixed effects, α_i, ensures that each indicator is estimated using only within firm variation in the dependent variable and time dummies, α_t controls for time trends. I recognize that incumbent supplier firms that previously manufactured reserved products may have secular time trends that differ from non-incumbents. To control for this, in various specifications, I include interaction between the year and incumbent dummies. The identification at the firm-level helps me absorb time-varying industry-level heterogeneity. Standard errors are corrected for heteroskedasticity and auto-correlation and clustered at the NIC 5-digit product level.

The removal of entry barriers should satisfy one requirement to be considered "exogenous": The occurrence of these events should be unrelated to an individual firm's strategic decision to extend trade credit. As discussed before, the reservation of products and their dereservation appear to be random in nature and varied across different industries.

In addition, I employ the Cox survival model to investigate whether any observable dif-

[20]Results are robust to alternative definitions of incumbent firms. Table 10, Column (5) reports results in which an incumbent is defined as a firm that made a reserved product three years before it was dereserved as its primary product.

[21]Since the coefficient on *Pre-Dereservation (-10)/Dereservation (+10)* represents the effect for more than one year, I do not report these coefficients.

ferences in average firm characteristics across industries predict the timing of dereservation. The sample for this regression spans 1990-2009 and includes the industries that were never reserved. In addition, I include the average characteristics of firms in each of the industries prior to dereservation. Specifically, I estimate industry-wide averages (at the NIC 5–digit level) of these firm characteristics over the period 1990-1996 which is six years before the first dereservation announcement. In alternative specifications, I include the average value of Accounts Receivables/Sales, Firm Size, Cash Holdings, Profitability, and Sales Growth. Table 2 reports the results from this analysis. I find that the coefficient on all these firm characteristics are insignificant for all the specifications, after including industry-fixed effects to control for industry-specific unobservables.

These results suggest that there is no systematic difference in the average characteristics between the firms in the early and the late dereserved industries. In further analysis below, I conduct various robustness checks and placebo tests to verify the validity of these results. Although, it is difficult to imagine some systematic and endogenous rationalization in this cross-section and time-wise pattern of reservation, I cannot entirely rule out this possibility.

Next, dereservation should relate to market competition in order to be a plausible variation for the threat of entry. Also, dereservation should not immediately affect the industry concentration. I use different measures of industry competition such as entry rate, product price, and market concentration index (HHI). I estimate *entry rate* as the ratio of the number of entrants to the number of incumbents in a 5-digit industry for a given year. For the *industry level product price index*, I create a firm-level price index for each product and take the weighted average over all the products in a given year for each 5–digit industry. For the firm-level price index, I use first observed quantity sold as a numeraire to track the price change. Finally, I measure market competition/concentration by the Herfindahl–Hirschman Index (HHI). I estimate a regression model similar to Equation (1) with (a) Entry Rate, (b) Log(Average Product Price Index), and (c) Log(HHI) (measured at 5-digit NIC product-level) as dependent variables. Here, the control group includes pre-event observations of reserved products that were dereserved later and products that were never reserved/dereserved products. Thus, a difference-in-differences estimate suggests how market competition is affected in the reserved product category compared to the control group.

Figure 2 plots the regression estimates and provides suggestive evidence for the nonexistence of pre-trends in entry rate/product prices/market concentration. Though not a perfect test, this shows that changes in competition did not really affect the dereservation decisions and therefore seem exogenous. The entry rate increases significantly after dereservation, and the average industry price index decreases by almost 18% four years after the dereservation. In addition, I find that market concentration decreases significantly by 10%, about 5-6 years after the dereservation. This suggests that competition within dereserved industries increased significantly subsequent to the removal of entry barriers. This is consistent with the fact that new entrants take time to set up plant and/or expand capacity to manufacture dereserved products. These new entrants may start manufacturing within a few years of dereservation, but they will start affecting market concentration only after 5-6 years as they begin to capture significant market share. This delayed effect of dereservation on market concentration helps me identify the *threat* of entry from *actual* entry. This suggests that dereservation had real effects on both product market competition and the threat of entry, thereby affecting firms' policies. In effect, dereservation seems to be a good instrument for the threat of entry.

4 Results

4.1 Main Results

I start my analysis in this section by plotting regression coefficients from my baseline specification, Equation (1) for dereserved products only.[22] Figure 3 shows the difference-in-differences estimate of trade credit granted (*accounts receivables over sales*) in event time and the 1% confidence interval around this difference. The evidence I present includes up to nine years after the dereservation event, but I am interested only in the narrow window of the first six years after dereservation, in which the *threat* of entry may dominate over the *actual* entry. As shown in Figure 2 (C) and in the discussions above, I find a significant change in market concentration only after five to six years. This short window of six years around the event, therefore, captures the average causal effect of the pre-emptive action.

The patterns in the figure are striking. First, there are no discernible pre-trends in my data: The difference between the treatment and control groups is statistically insignificant in the five

[22] Note that here the control group includes pre-event observations of firms producing reserved products that were dereserved later. It does not include never treated firms; so I do not include incumbent-year fixed effects.

years before the dereservation. Second, after the dereservation, there is a clear and statistically significant jump in trade credit granted (measured as accounts receivables over sales) one year after the dereservation. Also, the effect stabilizes after three years of dereservation. In further analysis, in tables I report results for coefficients on *Dereservation (k=-3)* and *Dereservation (k=-2)* and *Dereservation (k=0)* to *Dereservation (k=+6).*

Table 3, Column (1) presents the above results in tabular form. In Column (2), I include firms whose products were never reserved/dereserved, and I include an interaction between the year and the incumbent dummies (incumbent × year fixed effects). Similar to Column (1), in this specification I also find that the coefficients on *Dereservation (k=-3)* and *Dereservation (k=-2)* are statistically insignificant. This further confirms that the dereservation process is exogenous to a firm's strategic decision to change trade credit. I find that the coefficients on *Dereservation (k=0)* through *Dereservation (k=+6)* are positive and significant in Column (1) and Column (2). This demonstrates that after the dereservation of products, investment in accounts receivable is grows more rapidly than sales. The positive and significant coefficient of *Dereservation (k=+1)* implies that from one year before to one year after the removal of the entry barrier, the ratio of accounts receivable to sales for treated firms increases by 8.9 % (0.017 divided by the pre-dereservation mean of 0.19) relative to the ratio for the control firms.[23]

4.1.1 Placebo Dereservation

In order to alleviate concerns that my results may be driven by industry conditions, I employ a placebo test. This test would expect my results to remain even if I replaced my *Dereservation* dummy as one for firms manufacturing *similar* products which were never reserved/dereserved. Since these products were likely to face similar demand/supply shocks (but were reserved/deserved), any effects of a magnitude similar to the effects shown earlier would raise concerns about dereservation as a valid instrument for entry threats.

In this subsection I provide results from a randomized assignment of dereservation years to similar products, keeping the distribution of the event years unchanged. For each product dereserved (defined at the 5-digit level) in the sample, I create a pseudo–dereservation sample. In this sample I assign products to one randomly selected similar product (i.e., a product at the

[23]The observations in regressions do not match the summary statistics because *reghdfe* STATA command automatically drops singletons to correctly estimate the clustered standard errors (Correia, 2015). All the results are qualitatively similar if one keeps the singletons, and they are available on request.

same 4-digit level) instead of the product that was actually dereserved. For each of the pseudo–dereservation samples, I run a regression as in Table 3, Column (2) and I save the relevant coefficients. I repeat this procedure 5,000 times to obtain a (nonparametric) distribution of coefficients obtained from the placebo dereservation samples.

In Figure 4, I plot these distributions of the coefficients. The black line embedded in the graphs represents the regression coefficient obtained using the actual data. For example, Figure 4 shows that the coefficient for *Dereservation (k=+1)* in my placebo sample is located in the tails of the placebo distribution (p-value<.001). Thus, unobserved industry shocks that affect trade credit supply do not drive my results. This justifies my identifying assumption.

4.1.2 Exposure to Dereservation

As mentioned earlier, one of the advantages of using Indian data is the availability of sales data for each product. This helps me identify a firm's exposure to a threat of entry. I measure the exposure of firms to dereservation as proxied by the sales contribution of affected products to a firm's total sales. In general, firms produce more than one product to remain competitive in a market. In fact, Goldberg, Khandelwal, Pavcnik, and Topalova (2010) find that Indian multiproduct firms are quite similar to their counterparts in US manufacturing firms studied by Bernard, Redding, and Schott (2011). Like, Goldberg, Khandelwal, Pavcnik, and Topalova (2010), I find that the typical firm in my sample produces two products, and the average sales revenue from the primary product varies from 70%–95%.

I measure the exposure of incumbent firms to threats of entry using a proportion of its sales revenue from dereserved products in the pre-dereservation period. I report the results in Table 3, Column (3). I find that incumbent firms with greater exposure to threats of entry extend more trade credit. In addition, the ratio of accounts receivables to sales increases by 2 percentage points (an approximate 8.2% increase relative to the pre-dereservation mean ratio of 0.19, with 0.81 exposure: 0.020.81/0.19) four years after the dereservation of the firm's primary product. Column (4) is similar to Column (3), except Column (4) uses Log(1+Accounts Receivables) as a dependent variable. I find similar results.

4.1.3 Industry Unobservables

Next, one might be concerned that trade credit terms vary across industries over time, or one may be concerned that dereservation may be correlated with time-varying industry unobservables (e.g., demand/supply conditions). To mitigate these concerns, I control for unobserved heterogeneity at the industry level. In Table 3, Column (5), I carefully interact the 4-digit industry-year fixed effects for both the treatment (incumbents) and never-treated control group. The effect of dereservation remains positive and statistically significant. After including industry-year fixed effects at the 4-digit level, which is possible because dereservations are measured at the 5-digit level, I find that from one year before to one year after the removal of an entry barrier, the ratio of accounts receivable to sales of the treated firms increases by 2.4 pp relative to the ratio of control firms. This effect is economically large; the jump in trade credit represents a relative increase of about 10.2% from the pre-event level.

Table 3, Column(6) shows the results after including control variables from prior literature (Petersen and Rajan, 1997; Barrot, 2016). To ensure that the inclusion of the control variables does not bias estimates, I lag the control variables by a year. I include 1–year lagged value of firm size, cash holdings, profitability, and sales growth as controls. I find that the inclusion of the control variables has a negligible effect on the size of the coefficients of interest. For example, the coefficient on *Dereservation (k=+1)* changes from 0.024 in Column (5) of Table 3 to 0.026 in Column (6) of Table 3. This effect is economically large; the jump in trade credit represents a relative increase of about 11% from the pre-event level. In terms of number of days, the term increases by almost 7 days ($0.026 \times 0.81 \times 365$) with previous level of 69 days (0.19×365). A survey conducted for 141 Indian firms by the Center of Analytical Finance, Indian School of Business, finds that the typical discount for payment within 15 days is about 4% (De and Singh, 2013). The cost of trade credit for customer firm with 69 days (average term before dereservation) is about 27.03% ($0.04 \times 365/(69\text{-}15)$). With an increase in the term of credit by 7 days, one year after the dereservation, the cost of trade credit for a customer firm is about 23.93% ($0.04 \times 365/(76\text{-}15)$). This is equivalent to a price discount of 3.1%.[24]

These results are consistent with hypothesis *H1*: The strategic use of extending trade credit is to defend market share. With expected competition, incumbent firms extend more trade credit before the potential entrant starts manufacturing the product (Fisman and Raturi (2004), Hyndman and Serio (2010), and Barrot (2016)). This preemptive action to increase accounts

[24]The average term offered to customer by surveyed firms is about 62 days, which is quite similar to my sample firms i.e. 69 days.

receivables may serve as an entry barrier for a potential entrant.[25] Hence, the potential entrant requires not only technological and organizational expertise, but also the capacity to extend credit to the customer base.

4.1.4 Price of Output

As discussed above, incumbent firms extend more credit to their customers in order to defend their market share. But, do they simultaneously reduce prices or keep them unchanged? In my sample, I can observe the quantities sold and the sales revenue generated from each product for a given firm-year. Using this data, I create a firm-level product price index. First, I use the first observed quantity sold as a numeraire to track the price change. I identify the sample firm's primary product each year based on total sales revenue from its product portfolio.

Table 4 reports the results for difference-in-differences regressions for a sample firm's Primary Product Price Index. Column (1) shows the results with dereservation dummies and incumbent $\times$ 4-digit product $\times$ year fixed effects. The results are striking. The negative and significant coefficient of *Dereservation (k=+1)* implies that from one year before to one year after the removal of the entry barrier, the price of the treated firms' primary product decreases by 9.4 % relative to the control firms. While for next three years price differences were not significant for treated and control group. Interestingly, five years after the removal of the entry barrier we start to observe entrants gaining market share (i.e., in Figure 2 (C), log (HHI) is significantly lower after 5–6 years of dereservation). I find that prices fall by almost 17.5%. It appears that incumbent firms adjust prices in step-function. In Column (2), I replace dereservation dummies with the proportion of total sales revenue from de-reserved products in pre-dereservation period. Then, in Column (3) I add firm controls as in Table 3. Finally, in Columns (4)-(6), I adjust log(Primary Product Price Index) for inflation using a wholesale price index. The results are consistent: I find that firms do lower prices as a preemptive action. The average firm whose main product is dereserved, lowers prices by 8.6% (0.81×0.107) one year after the dereservation. After five years the price reduction is about 17.7% (0.81×0.219).

[25] I discuss the effectiveness of trade credit as a strategic tool in section 4.6

4.2 Access to Finance

It is important to understand whether a firm's choice of strategic action may be influenced by access to finance or financial strength. I test hypothesis *H2*: How an incumbent firm's response to a threat of entry varies with their financial strength or access to finance. I follow Hadlock and Pierce (2010) and use a firm size-age (SA) index as a proxy for financial strength. In addition, I use prior number of banking relationships as a proxy for access to finance and I report results in Table 5.

I re-estimate Column (6) of Table 3, my baseline specification, with interaction terms *Dereservation (k)* $\times$ *Low SA* and *Dereservation (k)* $\times$ *High SA,* where *Low SA* is a dummy variable that identifies firms with below-median Size-Age Index in the year prior to the dereservation of its primary product. I report the results in Panel A of Table 5. I do this for both *accounts receivables over sales* and *Log(Primary Product Price Index)* as dependent variables. The evidence suggests that firms with less financial constraint increase *accounts receivables over sales*, while firms with lesser financial strength cut prices. For above-median size-age index firms, one year after the dereservation, trade credit granted/sales increase by 0.061 (=0.8$\times$0.76), which is equivalent to 22 days (0.061$\times$365). The cost of trade credit with 73 days (the average term before dereservation for the above–median group) is about 25.17% (0.04$\times$365/(73-15)). With an increase in term of credit of 22 days, one year after the dereservation, the cost of trade credit for a customer firm is about 18.25% (0.04$\times$365/(95-15)). This is equivalent to a price discount of 6.92%. On the other hand, firms with less access to credit (i.e., firms with a below–median size-age index) lower prices by 28% (0.8$\times$0.355).

In addition to trade credit, the supply of credit from banks is an important source of finance for sample firms. For the median firm, 25% of assets are financed through bank credit. Also, the median firm in our data borrows from two banks before the dereservation of its main product.[26]

As in Panel A, Table 5, I re-estimate Column (6) of table 3 with *Less Banks* and *More Banks* as interaction terms, where *Less Banks* identifies firms with a below-median number of banking relationships one year prior to dereservation of its primary product. Table 5, Panel B show the results. I find that firms with more banking relationships at the time of the entry threat offered more credit to customers, while incumbent firms with less banking relationships

[26]The number of banking relationships is consistent with Gopalan, Mukherjee, and Singh (2016).

rely on lower prices.[27]

To further confirm the results, I use the establishment of debt recovery tribunals (DRTs) in India. These tribunals reduced the cost of legal enforcement of debt contracts in different states without any accompanying change in the underlying law. I use this as a proxy for access to finance. Various studies show that the lower enforcement costs are positively associated with the size of new loans and debt maturity, and they are negatively related to interest costs (Lilienfeld-Toal, Mookherjee, and Visaria (2012); Gopalan, Mukherjee, and Singh (2016); and Visaria (2009)). I report the results in the Internet Appendix Table IA5, Panel A.

I find that firms located in early DRT states were quick to respond to entry threats by increasing the credit offered to customers, while incumbent firms located in late DRT states relied on lower prices. I find that firms located in early DRT states increased trade credit by 18.9% (or an increase in the term of credit of 17 days above the previous level of 73 days) and, I find that these firms cut prices by 9.2% one year after the dereservation of its main product compared to one year before the dereservation. Firms in late DRT states do not change trade credit terms, and they cut prices by almost 40%.

I also use various other proxies for access to finance, such as relationships with large banks, access to bank credit using firm geographic location, and the supply of credit from product suppliers in the form of trade credit. I find consistent results. The results are reported in the Internet Appendix Table IA5, Panel B to Panel D.

Overall, the evidence supports the hypothesis that firms with better access to the financial market use trade credit as strategic tool, while other firms rely on limit-pricing. These findings are consistent with the large literature that indicates that deep-pocketed firms attempt to drive financially constrained competitors out of business (Telser, 1966; Bolton and Scharfstein, 1990), while here we identify the exact tool for this strategy: offering more credit.

4.3 Financing Trade Credit

In the last subsection, I find that firms with better access to financial markets use trade credit as a strategic tool. Its important to understand how firms finance this excess credit offered to customers.

[27] In this analysis, the number of observations decreases due to the non-availability of banking relationship data.

I estimate Column (6) of Table 3 for different dependent variables. Column (1) displays the results with (a) Trade Credit Received (scaled by assets), while Columns (2)-(8) report results with (b) Bank Credit (scaled by assets). In Columns (3)-(8), I report the results of regressions that estimate the response of incumbent firms to threats of entry based on a firm's access to credit, using the size-age index or the prior number of banking relationships as a proxy for financial strength or access to finance. For affected firms, I find trade payable (i.e., credit from supplier) appears to increase, but there is no change in bank credit. When I partition this based on a firm's size-age index and prior number of banking relationships, I find no difference between the two groups in regard to trade payable, so I do not report this here. Interestingly, I find that large-old firms and firms with more banking relationships receive more bank credit after dereservation. These results further validate the use of the size-age index and prior number of banking relationships as proxies for access to finance.

4.4 Demand for Trade Credit

In this subsection, I test hypothesis *H3*: how an incumbent's response to a threat of entry varies with the financial strength of its customers. In my data, I cannot observe the customers of a given firm, so I attempt to test this hypothesis by considering the sample firms as customers. I test whether incumbent firms whose main input is dereserved (i.e., the input market becomes more competitive) receive more credit or lower prices or both. Also, I test how access to finance affects these results.

4.4.1 Trade Credit Received

Table 7, Panel A shows the results for trade credit received as a dependent variable. I measure trade credit received as *accounts payable/total assets*. Column (1) shows results when the control group includes only pre-event observations of firms using reserved products that were dereserved later (incumbents). In Column (2) I include firms whose primary inputs were never reserved or dereserved, and I include an interaction between the year and incumbent dummies (incumbent $\times$ year fixed effects). Results from both columns suggest that incumbent firms whose main input is dereserved receive more more credit from their suppliers. The positive and significant coefficient of *Dereservation (k=+1)* implies that from one year before to one year after the removal of an entry barrier in a firm's main input market, the ratio of accounts payable to assets of the treated firms increases by 8.8 % (0.016 divided by the pre-dereservation mean of 0.18) relative to the ratio of control firms.

In Column (3), to estimate the exposure of a firm to dereservation, I replace dereservation dummies with the proportion of total raw material expenditure from dereserved products in the pre-dereservation period. Column (4) is similar to Column (3), except with Log(1+Accounts Payable) as a dependent variable. Column (5) is similar to Column (3), except with Accounts Payable scaled by Cost of Goods Sold as a dependent variable. Column (6) shows the results when I further add to Column(3) the 1-year lagged value of firm size (measured as log sales), cash holdings, profitability, and sales growth as controls. All regressions except Column (1) are with incumbent $\times$ 4-digit product $\times$ year fixed effects.

Overall, the evidence suggests that incumbent firms receive more credit from their suppliers after the dereservation of its main input.

4.4.2 Price of Inputs

Similar to price of output, in Table 4, I create a primary input price index and follow the same specifications. For inputs, I find that the price index for firm's primary inputs goes down by 11.3% to 12.9% in different specifications. The negative and significant coefficient of *Dereservation (k=+1)* in Column (6) of Table 7, Panel B, implies that from one year before to one year after the removal of an entry barrier in a firm's main input market, the Primary Input Price Index of treated firms decreases by 11.3 % (with an exposure ratio of 0.71 for the main input, so the effect is about 0.71×0.157) relative to the ratio of control firms.

4.4.3 Customer's Access to Finance

It is important to understand whether results vary by access to finance or the financial strength of customer firm. I test hypothesis *H3*: how an incumbent firm's response to a threat of entry varies with a customer's financial strength or access to finance. As in Table 5, here I partition the above results based on financial strength proxies (i.e., Size-Age index). Table 7, Panel C shows the results. These results suggest that firms with financial constraints (i.e., Low SA firms) receive more credit, while unconstrained firms accept price discounts. In Internet Appendix Table IA5, Panel E, I report results with DRT as proxy for financial constraints. I find that firms located in states with late DRT receive more trade credit.

The overall findings in this subsection suggest that financial constraints do impact the choice of strategic tool when firms face an entry threat from new competitors. Firms with deeper pockets offer more credit, and firms with financial constraints take more credit, when bargaining power shifts from the supplier to the customer. And, unconstrained customer firms demand lower prices when their suppliers face more competition.

4.5 Other Protectionist Policies

In India, the reservation of products to be manufactured only by small firms is not the only policy used to protect small firms. In addition to protection from large manufacturing units, small firms in India receive priority for bank credit. All banks (public and private) are required to lend at least 40% of their net credit to the "priority sector," which includes agriculture, agricultural processing, transport industry, and small–scale industry (SSI). If banks do not satisfy the priority sector target, they are required to lend money to specific government agencies at very low rates of interest. If they lend to the priority sector, they lend at the same interest rate as in other sectors (Banerjee and Duflo, 2014).

During my sample period, the definition of *small firms* for priority sector lending changed three times. I use this heterogenity to identify how incumbent firms in the priority sector react to a threat of entry. In January 1998, there was a change in the definition of the small scale industry sector. Before this date, only firms with a total investment in plant and machinery below INR 6.5 million were included. The reform extended the definition to include firms with investment in plants and machinery up to INR 30 million. In January 2000, the reform was partially undone by a new change: Firms with investment in plants and machinery between Rs. 10 million and INR 30 million were excluded from the priority sector. Later, with the MSME Act of 2006, the definition further changed so that firms with investment in plants and machinery less than INR 0.25 million were categorized as *micro enterprise* and those between Rs. 0.25 million and INR 50 million were categorized as *small enterprise.* The priority sector includes both micro and small enterprises.

I re-estimate Column (6) of Table 3, my baseline specification, with interaction terms *Dereservation (k)* $\times$ *PSL* and *Dereservation (k)* $\times$ *Non-PSL,* where *PSL* is a dummy variable that identifies firms in the priority sector in the year before the dereservation of its primary product, and I report the results in Table 8. I do this for both *accounts receivables over sales* and *Log(Primary Product Price Index)* as dependent variables. In my sample, 12% of firms that manufacture the reserved/dereserved products were in the priority sector, while the rest of the incumbent firms were capacity–constrained medium– or large–sized firms.[28]

The evidence suggests that small firms in the priority sector not only reduce prices but also offer trade credit to defend their market share.

4.6 Entry Deterrence

Finally, in this subsection, I test the hypothesis about whether firms that tried to defend their market share by increasing trade credit were successful. Note that these tests are only suggestive and cannot be interpreted as a causal relation. I observe the entry rates in 5-digit industries separately for firms that chose to increase trade credit after dereservation and those that do not. I report the results in Table 9.

I measure the entry rate as the ratio of the number of entrants to the number of incumbents in 5-digit industry for a given year. Then, I observe the change in entry rates after dereservation for the group of firms that have increased (decreased) trade credit in the same year as the dereservation and one year after. I find that entry rates have increased after the dereservation for both groups, but the increase is lower for the group that chose to increase trade credit in the same year as the dereservation and one year after. For Dereservation(k=+2), the coefficient of -0.218 in Column (3) implies deterrence of one firm in an industry with five incumbents. Overall, these results suggest that the entry of new firms after dereservation is negatively correlated with the trade credit terms in the industry.

The firms that managed to enter tend to match the trade credit terms with incumbent firms. Figure 5 plots the average of trade credit granted for incumbent firms and entrants after the dereservation. Two years after the dereservation, I find the difference between the average trade credit granted/sales for incumbent firms and entrants to be insignificant.

5 Robustness

In this section, I consider various refinements of, and address potential concerns with, my baseline specification (Table 3, Column (6)). I report the findings in Table 10. Panel A shows the results with *accounts receivables over sales* as a dependent variable, while Panel B shows

[28] As mentioned before, my sample may not be capture very small and micro enterprises that were manufacturing the reserved products, but the data capture relatively medium– and large–sized incumbent firms that were capacity constrained.

the results for *Log(Primary Product Price Index)*.

(i) *Potential Confounding Factors*

As discussed earlier in the data section, major changes in policies regarding foreign investment occurred in the early 1990s and then stalled during the period of dereservation reform. Nataraj (2011) shows that tariffs were largely harmonized across industries by the late 1990s. Although some reductions occured during the 2000s, the variation in tariff rates across product types had already fallen dramatically by the start of 1997. I estimate Column (6) of Table 3 for the sample beginning from 1997, and I report the result in Column (1) of Table 10. The effect of the removal of entry barrier on trade credit remains statistically and economically significant.

(ii) *Sub-period Analysis*

As most of the products in the sample were dereserved after 2003, Column (2) shows the results for the sample beginning from 2003. The results are robust to this sample selection.

(iii) *Excluding Firms Manufacturing Chemical Products*

In Column (3), I exclude firms that manufacture chemical products, an industry with many reserved/dereserved products, and I find that the results remain significant.

(iv) *State-Year Fixed Effects*

Various state-specific economic conditions and other political factors may be correlated with the special committee's decisions rather than the product-level characteristics. I include state-specific year fixed effects to control for such unobservables (Column (4)). The results reveal a similar pattern with these modifications.

(v) *Alternative Definition of Incumbent Firms*

In Column (5), I report the results after changing the definition of incumbent firms. I redefine the *incumbent* as a firm that manufactured a reserved product as its primary product three years before it was dereserved. The results are robust and quantitatively similar to the baseline results.

(vi) *Exclude Industries with Large Firms*

Large incumbent firms that are market leaders, in order to protect their market power, may try to influence the decision of policy makers in the country. Therefore, the timing of dereservation may not be completely exogenous to firm policies in those industries or products. In Column (6), I exclude from the treatment group the industries (at the 5-digit) with large leader firms.

(vii) *Firm-level clusters*

Finally, in Column (7), I correct standard errors for clusters at the firm level instead of at the NIC 5-digit product level. The results are robust to these changes.

Overall, I find that my basic result is robust: Given an increased threat of entry, incumbent firms increase payment terms and cut prices.

6 Conclusion

This study examines how financial constraints affect incumbent supplier firm's choice of extending more trade credit versus offering price discounts when facing an increased threat of entry from competitors (distinct from actual entry), by using trade credit or price as a credible preemptive strategy to deter potential entrants. To test this hypothesis and its implications, I exploit, *plausibly* exogenous, staggered removal of product–level entry barriers over time for Indian manufacturing firms. This reform triggered an increase in competition that reduced market concentration by 10% in affected industries over the subsequent five to six years. In a difference-in-differences setting, I find that an average incumbent firm extends 11% more trade credit and lowers prices by 8.6% when the threat of entry increases. Interestingly, firms with deeper pockets offer longer terms on trade credit, while financially constrained firms rely on price discounts. Also, I find that financially constrained customers accept credit offers while unconstrained customers accept price cuts. I further confirm my results using various proxies for financial constraints and national policy changes that improved access to financing. This paper thus complements recent articles on trade credit and industry structure, thereby extending the literature on trade credit regarding its role as a strategic tool to deter future entry.

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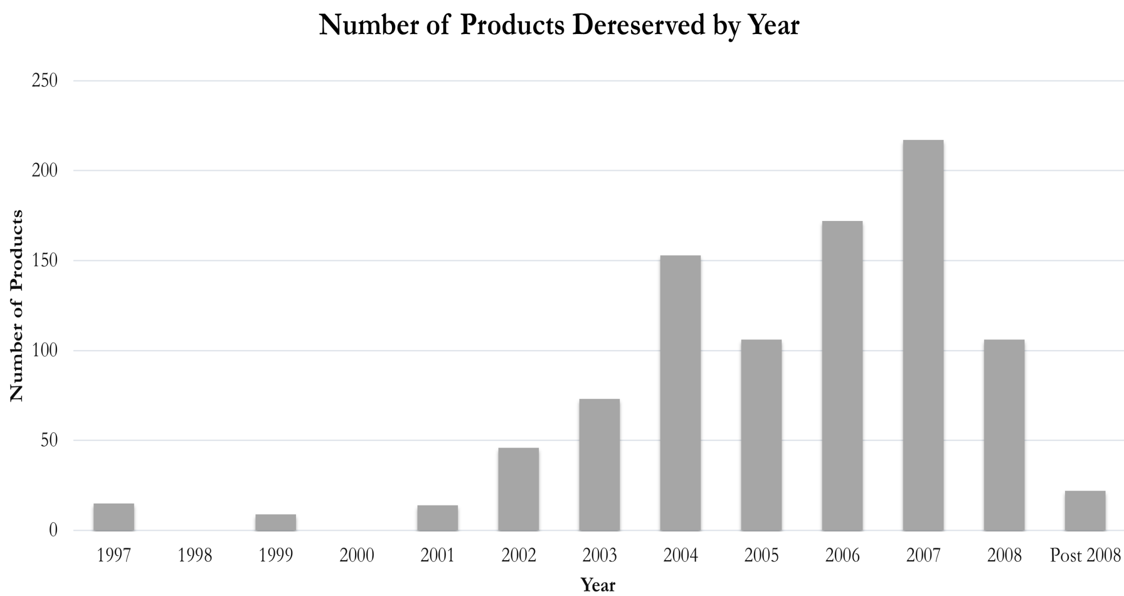
Figure 1: Timing of Dereservation Reform

The figures plot the number of products dereserved by the Government of India over the years in different industries. Figure 1A plots the time series from 1997 onward. Figure 1B plots the number of products dereserved in each industry over the same period.

1A: Year-Wise



1B: Industry-Wise

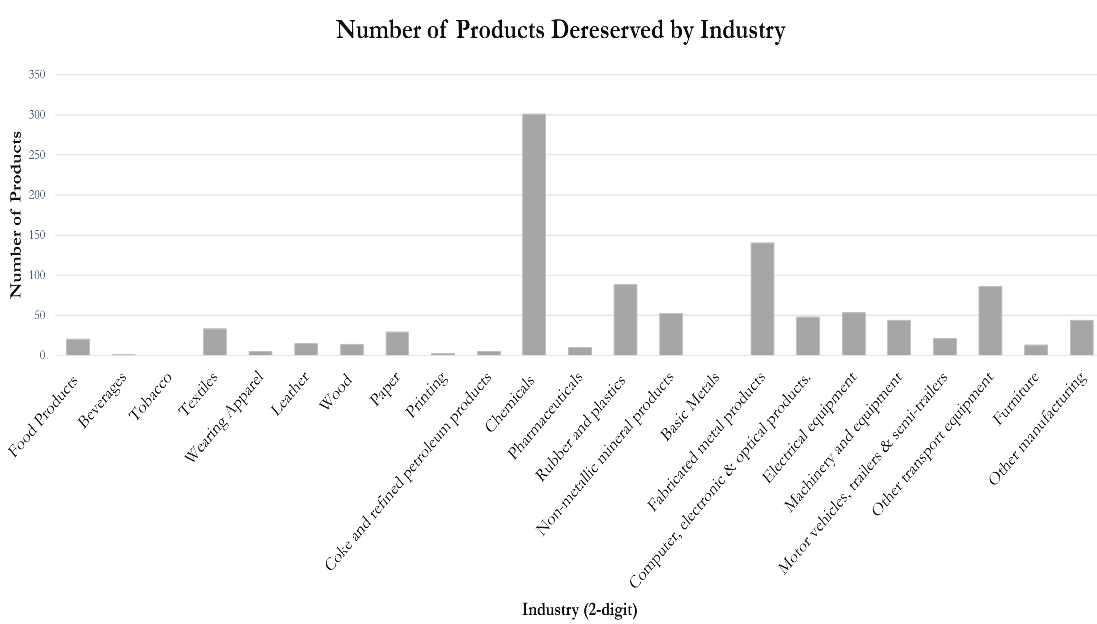
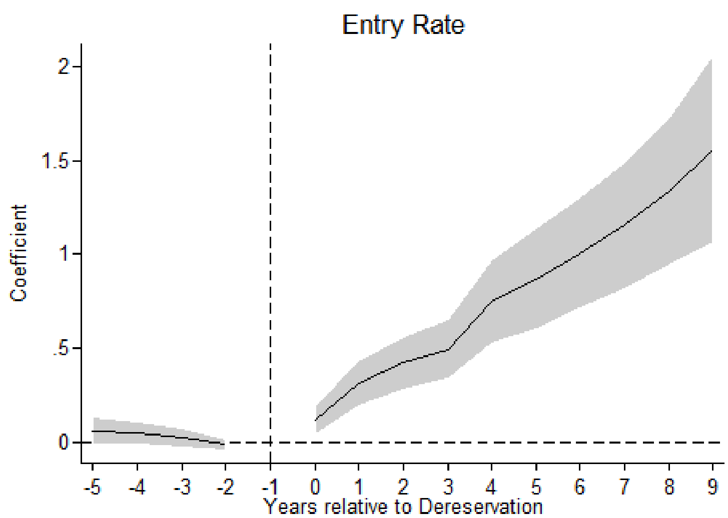


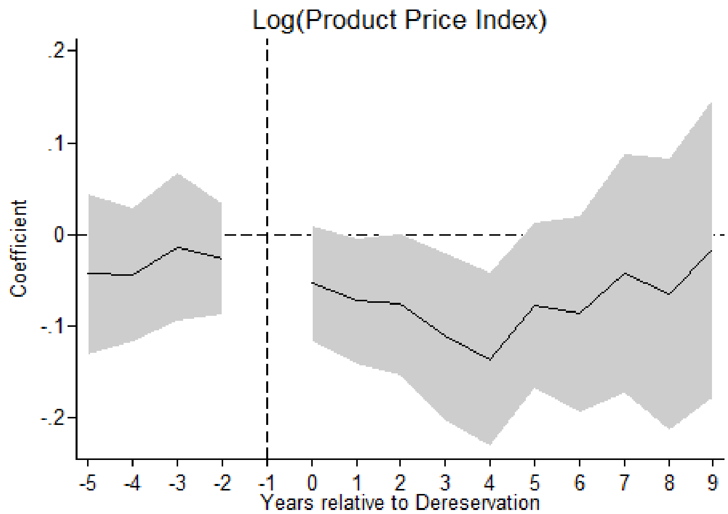
Figure 2: Effect of Dereservation on Industry Competition

The figures show how dereservation relates to industry competition (i.e., how entry rate, product prices, and market concentration index (HHI) are affected in a reserved product category compared to a control group). The control group includes (a) pre-event observations of firms producing reserved products that were dereserved later and (b) firms whose products were never reserved or dereserved. I estimate a regression model similar to Equation (1) with (a) Entry Rate (i.e., number of new entrants scaled by incumbents), (b) Average Product Price Index (see Section 3) and (c) Log(HHI) (measured at the 5-digit NIC product-level) as dependent variables. The model also includes 5-digit product-level fixed effects and year fixed effects. Here, I plot the estimated coefficients for Pre-Dereservation (k=-5 to k=-2) to Dereservation (k=0 to k=+9), along with 1% confidence intervals around this difference.

2A: Entry Rate



2B: Product Price Index



2C: Market Concentration

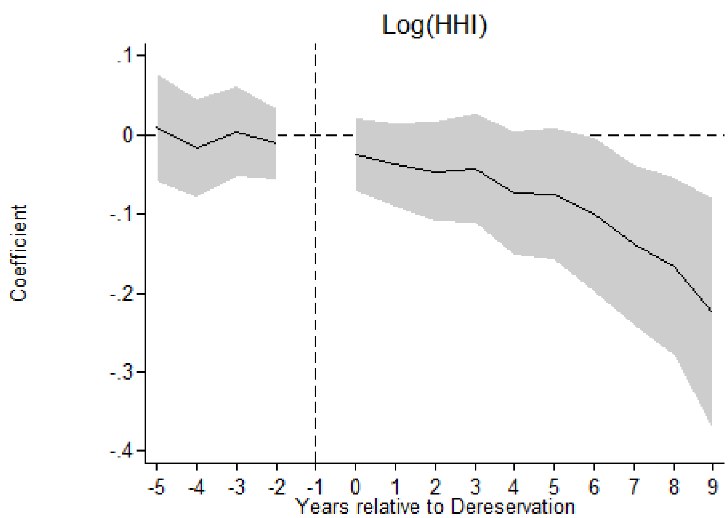


Figure 3: Effect of Dereservation on Trade Credit

The figure below plots the changes in mean trade credit granted (measured as accounts receivables scaled by sales) following dereservation. The figure shows how incumbent firms respond to dereservation within the reserved product category. I estimate Equation (1) for dereserved products and plot the estimated coefficients from Pre-Dereservation (k=-5 to k=-2) to Dereservation (k=0 to k=+9) along with 1% confidence intervals around this difference.

Difference-in-Differences

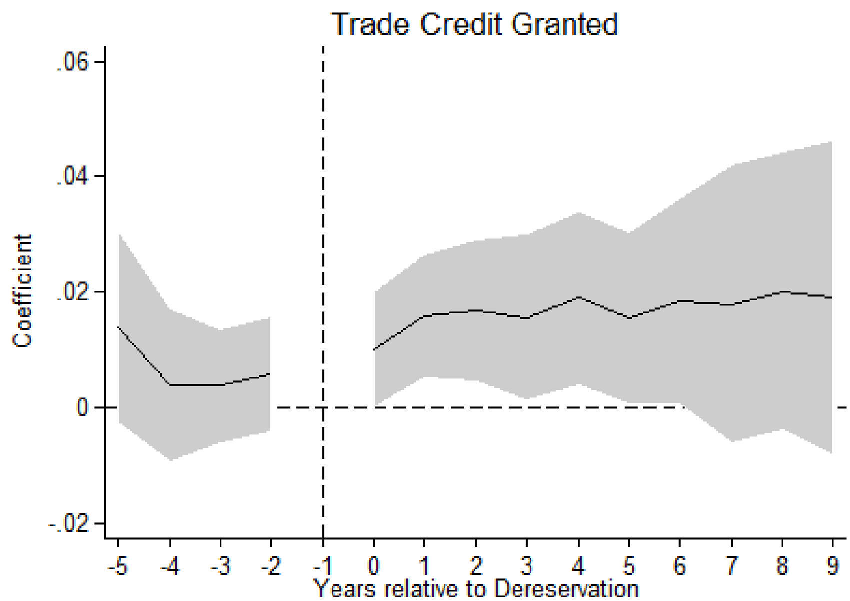


Figure 4: Non-Parametric Distribution of Placebo Dereservation Estimates

The figure below provides the result of the randomized removal of entry barriers of similar products, keeping the distribution of the event years unchanged. For each product dereserved (defined at the 5-digit level) in the sample, I create a pseudo–dereservation sample in which, instead of the product that was actually dereserved, I assign it to one randomly selected similar product(i.e., at the same 4-digit level). In each of the pseudo–dereservation samples, I run my baseline regression as in Table 3, Column (2), and save the relevant coefficients. Here, I plot the distribution of the coefficients. The black line embedded in the graph represents the regression coefficient obtained using the actual dereservation of products in the first year after the dereservation.

Placebo

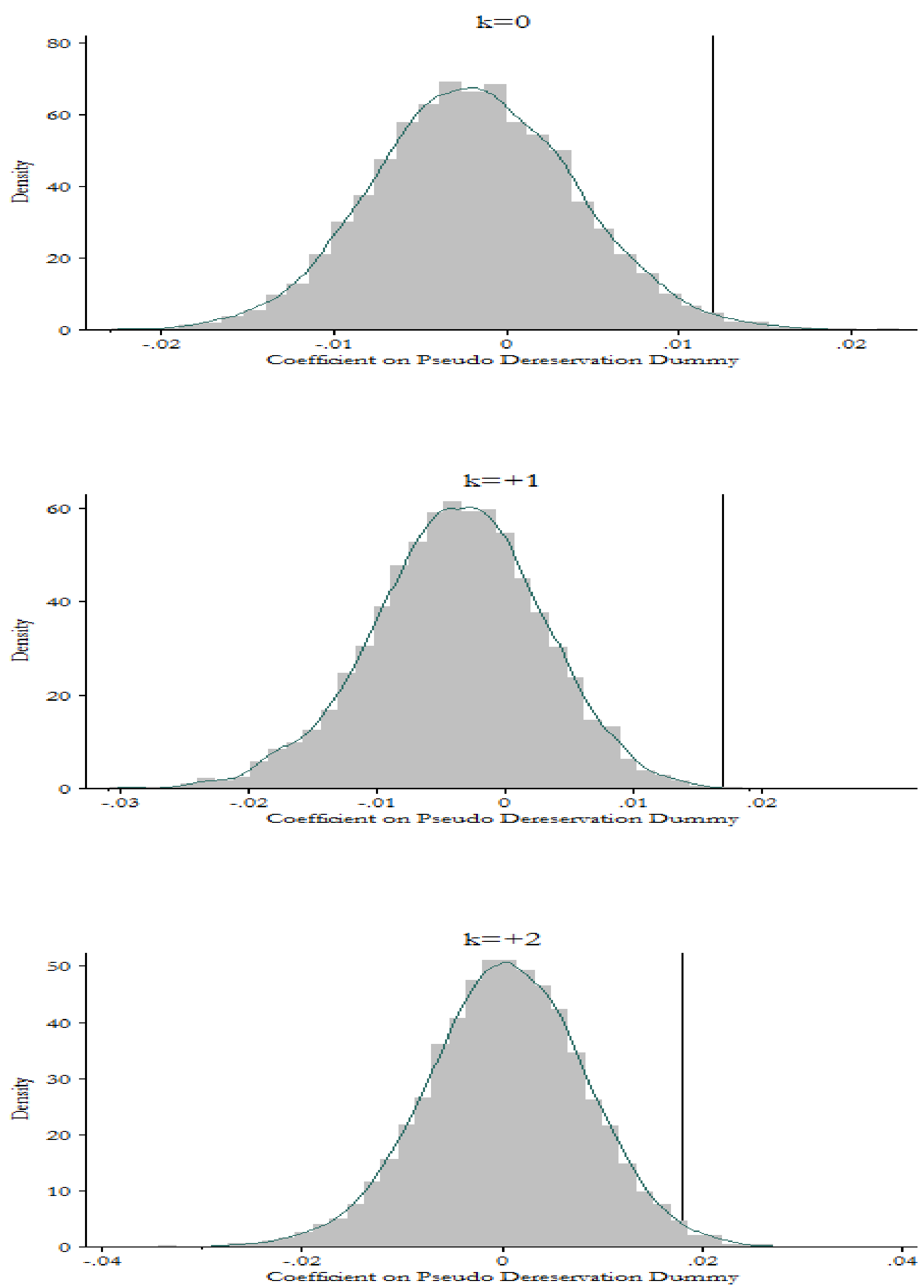


Figure 5: Ex Post: Trade Credit Granted

The figure below plots the average trade credit credit granted (scaled by total sales) for both incumbents firms and new entrants after the dereservation.

Trade Credit Granted: Incumbents vs. Entrants

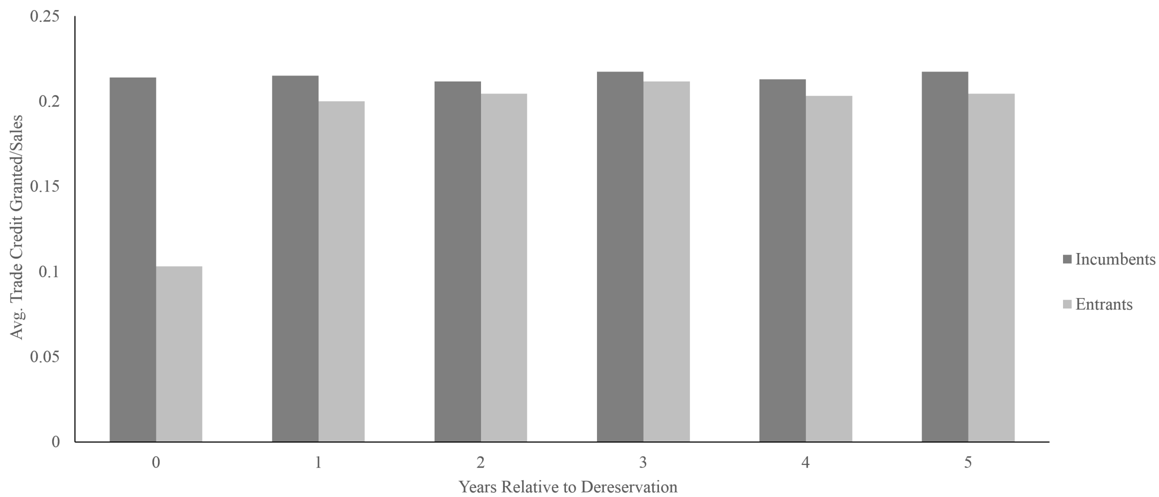


Table 1: Summary Statistics

This table reports descriptive statistics for my sample firms. I compile a firm-level panel data set based on the Prowess database, collected by the Centre for Monitoring the Indian Economy (CMIE). The database contains information primarily from the income statements and balance sheets of about 34,197 publicly listed/private companies, and almost 10,724 of them are in the manufacturing sector. Indian firms are required by the 1956 Companies Act to disclose product-level information on capacities, production, and sales in their annual reports. The CMIE compiles these detailed quantitative data, which makes it possible to track a firm's products over time. Furthermore, for each product manufactured by the firm, the data set provides the value of sales, quantity, and units. The Prowess database is therefore particularly well suited for understanding how firms adjust their financial policies when their product lines are under threat of competition. The reservation policy was meant only for manufacturing firms in India; I keep only non-services and non-government firms in my sample. All the financial variables are inflation adjusted using WPI at 2004-2005 prices. The data also correct for changes in the financial reporting year by adjusting values for number of months. To make sure that results are not affected by outliers, I winsorise all the ratios at 1% and drop observations that have a ratio of accounts receivables to sales greater than 1. The data period spans from the beginning of the year 1990 to the end of 2013.

Variables	Definition	Obs	Mean	Stdev.	Median
Trade Credit Granted	$\frac{\text{Accounts Receivables}}{\text{Sales}}$	32,603	0.19	0.15	0.16
Price of Output	Log of Primary Product Price Index	27,609	0.46	0.32	0.67
Trade Credit Received	$\frac{\text{Accounts Payables}}{\text{Assets}}$	40,188	0.18	0.16	0.13
Price of Input	Log of Primary Raw Material Price Index	32,454	0.24	0.25	0.61
Bank Credit	$\frac{\text{Bank Borrowing}}{\text{Total Assets}}$	31,354	0.28	0.22	0.25
Firm Size	Total Assets(in $ Mn.)	32,603	53.2	200.6	11.1
Cash Holdings	$\frac{\text{Cash}}{\text{Total Assets}}$	32,603	0.06	0.09	0.03
Profitability	$\frac{\text{EBITDA}}{\text{Total Assets}}$	32,603	0.12	0.10	0.12
Sales Growth	$\frac{\text{Sales}-\text{Lagged Sales}}{\text{Lagged Sales}}$	32,603	0.14	0.65	0.06

Table 2: Effect of Firm characteristics on Timing of Dereservation

This table reports results from Cox survival regressions that investigate the timing of the dereservation of different products. The table looks at industry-level average firm characteristics. Firm characteristics are calculated as industry averages (at the NIC 5-digit) from 1990-1996 which is six years before the announcement of first product dereservation. All the specifications include industry fixed effects at the 4-digit level. Standard errors are corrected for heteroskedasticity and autocorrelation, clustered at NIC 5-digit product level, and reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1% respectively. All the variables are winsorized at 1%.

	Time to Dereservation					
	(1)	(2)	(3)	(4)	(5)	(6)
Trade Credit Granted(Industry Avg.)	-1.240					3.683
	(1.978)					(3.826)
Firm Size (Industry Avg.)		.251				.387
		(.312)				(.286)
Profitability(Industry Avg.)			4.588			4.267
			(4.875)			(3.685)
Cash Holdings(Industry Avg.)				.988		1.181
				(2.296)		(3.319)
Sales Growth(Industry Avg.)					1.330	2.071
					(1.142)	(1.307)
Industry Fixed Effects	4-digit	4-digit	4-digit	4-digit	4-digit	4-digit
Obs.	1065	1065	1065	1065	1065	1065
Log Pseudo likelihood	-241.64	-241.309	-241.303	-241.692	-240.477	-238.875
χ^2 statistic	16884.3	12924.58	1887.563	56474.67	1892.95	16223.03

Table 3: Trade Credit Extended

This table presents the estimates from difference-in-differences regressions for *accounts receivables over sales*, a proxy for payment terms (except Column (4)), around the removal of entry barriers (dereservation of products). The estimates are based on the following regression equation:

$$\mathrm{y}_{it} = \alpha_0 + \sum_{k=-2}^{-10} \beta_k Pre - Dereservation(k)_{it} + \sum_{k=0}^{10} \beta_k Dereservation(k)_{it} + \alpha_i + \alpha_t + \varepsilon_{it}. \tag{1}$$

I define treatment as the elimination of small-scale reservation law on the incumbent firm's primary product in the NIC 5-digit category. A firm's primary product is based on maximum sales revenue in the year preceding dereservation. *Pre-Dereservation(k)$_{it}$* (*Dereservation(k)$_{it}$*) is a dummy variable that takes a value one if it is k years before (after) the incumbent firm's primary reserved product has been dereserved, and takes a value of zero otherwise. An *incumbent* firm is defined as a firm that made a reserved product as its primary product before it was dereserved. *Dereservation(k)$_{it}$* is a dummy that identifies the group of firms under threat of entry (or exposure of the firm to entry threat by using the proportion of total sales revenue from the dereserved product in pre-dereservation period.) The model is fully saturated, with the year immediately before the dereservation as the excluded category. Therefore, the coefficients on Pre-Dereservation(k)$_{it}$ (Dereservation(k)$_{it}$) compare the level of the dependent variable k years before (after) the dereservation to the year immediately before its de-reservation. The inclusion of firm fixed effects, α_i, ensures that each indicator is estimated using only within-firm variation in the dependent variable. Time dummies, α_t, control for country-level trends. In Column (1), control group includes only pre-event observations of firms producing reserved products that were dereserved later (incumbents). In Column (2) I include firms whose products were never reserved/dereserved and include an interaction between the year and incumbent dummies (incumbent $\times$ year fixed effects). In Column (3), to estimate exposure of the firm to an entry threat, I replace dereservation dummies with the proportion of total sales revenue from the dereserved product in pre-dereservation period. Column (4) is similar to Column (3) except with Log(1+Accounts Receivables) as a dependent variable. Column (5) is similar to Column (3) in which I replace *incumbent* $\times$ *year* fixed effects with *incumbent* $\times$ *4-digit product* $\times$ *year* fixed effects. To Column(5) I further add the *one year lagged* value of firm size (measured as log sales), cash holdings, profitability and sales growth as controls, and Column (6) shows the results. In this paper, I present estimated coefficients for Pre-Dereservation (k=-3 to k=-2) to Dereservation (k=0 to k=+6). Standard errors are corrected for heteroskedasticity and autocorrelation, clustered at NIC 5-digit product level, and reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1%, respectively. All the variables are winsorized at 1%.

	Trade Credit Granted (*Accounts Receivables over Sales*)					
	Only Treatment Firms	All Incumbents	Exposure to threat	Logged	Product Year FEs	Firm Controls
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=-3)	0.004	0.004	0.006	0.079	-0.002	0.004
	(0.005)	(0.005)	(0.006)	(0.060)	(0.010)	(0.010)
Dereservation(k=-2)	0.006	0.007	0.009	0.083	0.014	0.016
	(0.005)	(0.005)	(0.007)	(0.055)	(0.012)	(0.012)
Dereservation(k=0)	0.010^{**}	0.012^{**}	0.013^{**}	0.134^{**}	0.011	0.015
	(0.005)	(0.005)	(0.006)	(0.054)	(0.010)	(0.011)
Dereservation(k=+1)	0.016^{***}	0.017^{***}	0.019^{***}	0.129^{**}	0.024^{**}	0.026^{**}
	(0.005)	(0.006)	(0.007)	(0.051)	(0.012)	(0.012)
Dereservation(k=+2)	0.017^{***}	0.019^{***}	0.019^{***}	0.195^{***}	0.020^{*}	0.022^{**}
	(0.006)	(0.006)	(0.007)	(0.055)	(0.011)	(0.011)
Dereservation(k=+3)	0.016^{**}	0.017^{**}	0.017^{**}	0.208^{***}	0.031^{**}	0.034^{**}
	(0.007)	(0.007)	(0.009)	(0.063)	(0.015)	(0.015)
Dereservation(k=+4)	0.019^{**}	0.022^{***}	0.020^{**}	0.219^{***}	0.026^{*}	0.031^{**}
	(0.008)	(0.007)	(0.009)	(0.067)	(0.016)	(0.015)
Dereservation(k=+5)	0.015^{**}	0.016^{**}	0.015^{*}	0.219^{***}	0.029^{*}	0.034^{**}
	(0.008)	(0.007)	(0.009)	(0.079)	(0.017)	(0.017)
Dereservation(k=+6)	0.018^{**}	0.020^{**}	0.017^{*}	0.204^{**}	0.024	0.029
	(0.009)	(0.008)	(0.010)	(0.081)	(0.019)	(0.019)
Firm FEs	✓	✓	✓	✓	✓	✓
Year FEs	✓					
Incumbent × Year FEs		✓	✓	✓		
Incumbent × 4-digit Product×Year FEs					✓	✓
Firm Controls						✓
Obs.	13,989	31,918	31,918	31,918	31,161	31,161
R^2	0.634	0.652	0.652	0.874	0.701	0.709

Table 4: Price of Output

This table presents the estimates from difference-in-differences regressions for a sample firms's Primary Product Price Index around the time of the removal of entry barriers (dereservation of products). I re-estimate Equation (1) with log(Primary Product Price Index). Column (1) shows the results with dereservation dummies and *incumbent* × *4-digit product* × *year* fixed effects. In Column (2), I replace dereservation dummies with the proportion of total sales revenue from the dereserved product in the pre-dereservation period. In column (3), I add firm controls, as in Table 3. In Columns (4)-(6), I adjust log(Primary Product Price Index) for inflation using the wholesale price index. In this table, I present estimated coefficients for Pre-Dereservation (k=-3 to k=-2) to Dereservation (k=0 to k=+6). Standard errors are corrected for heteroskedasticity and autocorrelation, clustered at NIC 5-digit product level, and reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1%, respectively. All the variables are winsorized at 1%.

	Log(Primary Product Price Index)					
				Adjusted for Inflation		
	Dummies	Exposure to threat	Firm Controls	Dummies	Exposure to threat	Firm Controls
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=-3)	-0.003	-0.024	-0.029	-0.006	-0.026	-0.031
	(0.040)	(0.044)	(0.044)	(0.041)	(0.045)	(0.045)
Dereservation(k=-2)	-0.043	-0.055	-0.058	-0.040	-0.054	-0.056
	(0.033)	(0.044)	(0.044)	(0.032)	(0.043)	(0.044)
Dereservation(k=0)	-0.057	-0.069	-0.073	-0.060	-0.076	-0.080*
	(0.040)	(0.048)	(0.048)	(0.040)	(0.047)	(0.047)
Dereservation(k=+1)	-0.094***	-0.101***	-0.105***	-0.092***	-0.102**	-0.107***
	(0.031)	(0.039)	(0.039)	(0.032)	(0.040)	(0.040)
Dereservation(k=+2)	-0.068	-0.072	-0.077	-0.063	-0.074	-0.079
	(0.049)	(0.058)	(0.058)	(0.049)	(0.059)	(0.060)
Dereservation(k=+3)	-0.054	-0.057	-0.063	-0.040	-0.053	-0.060
	(0.056)	(0.073)	(0.074)	(0.056)	(0.074)	(0.075)
Dereservation(k=+4)	-0.072	-0.070	-0.078	-0.066	-0.075	-0.084
	(0.052)	(0.075)	(0.076)	(0.049)	(0.072)	(0.072)
Dereservation(k=+5)	-0.175**	-0.201**	-0.211**	-0.162**	-0.209**	-0.219**
	(0.071)	(0.091)	(0.091)	(0.066)	(0.087)	(0.087)
Dereservation(k=+6)	-0.094	-0.049	-0.059	-0.093	-0.072	-0.083
	(0.093)	(0.123)	(0.124)	(0.086)	(0.115)	(0.116)
Firm FEs	✓	✓	✓	✓	✓	✓
Incumbent × 4-digit Product×Year FEs	✓	✓	✓	✓	✓	✓
Firm Controls			✓			✓
Obs.	23,506	23,506	23,506	23,506	23,506	23,506
R^2	0.74	0.74	0.741	0.788	0.788	0.788

Table 5: Access to Finance

This table reports the results of regressions that estimate the differential response of incumbent firms to a threat of entry based on their financial strength/access to finance. I use size-age (SA) index (Hadlock and Pierce, 2010) and prior number of banking relationships as a proxy for financial strength/access to finance. Panel A shows the results of using the size–age index. Panel B shows the results of using the number of banking relationships. For size-age index/banking relationships, I estimate Column (6) of Table 3 after replacing each *Dereservation(k)* (and *Pre-Dereservation(-k)*) with the interaction terms *Dereservation (k)* × *Low SA/Less Banks* and *Dereservation (k)* × *High SA/More Banks*, where *Low SA* is a dummy variable that identifies firms with below-median *Size-Age* Index in the year before the dereservation of its primary product. *Less Banks* identifies the firms with below-median *banking relationships* in the year before the dereservation of their primary product. In this specification, I also include a full set of interaction terms between different groups for access to finance and time fixed effects. This allows two groups of firms to have a differential time trend. Also, I control for industry–specific time fixed effects at the NIC 4-digit level, and I add firm controls for each group separately. In the column titled *Diff*, I test whether the coefficients estimated for different groups are significantly different. Note that in this test, my sample is confined to treated firms that exist one year before the dereservation. Columns (1)-(3) report the results with *Accounts Receivables over Sales* as a dependent variable, while Columns (4)-(6) report the results with *Log(Primary Product Price Index)*, adjusted for inflation, as a dependent variable. Standard errors are corrected for heteroskedasticity and autocorrelation, clustered at NIC 5-digit product level, and reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1%, respectively.

Panel A: Size–Age Index

	Trade Credit Granted			*Log(Primary Product Price Index)*		
	Low SA	High SA	Diff (2)-(1)	Low SA	High SA	Diff (4)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	-0.000	0.051	0.051	-0.339	-0.057	-0.282
	(0.021)	(0.035)	(0.037)	(0.225)	(0.045)	(0.224)
Dereservation(k=+1)	-0.001	0.076**	0.077**	-0.355*	-0.060	-0.295
	(0.018)	(0.036)	(0.037)	(0.213)	(0.049)	(0.213)
Dereservation(k=+2)	0.004	0.072**	0.068*	-0.400*	-0.021	-0.379
	(0.024)	(0.036)	(0.041)	(0.238)	(0.070)	(0.251)
Dereservation(k=+3)	0.006	0.068	0.063	-0.349	-0.013	-0.336
	(0.027)	(0.042)	(0.050)	(0.300)	(0.087)	(0.298)
Dereservation(k=+4)	0.005	0.059	0.054	-0.319	-0.052	-0.267
	(0.028)	(0.039)	(0.043)	(0.268)	(0.114)	(0.276)
Dereservation(k=+5)	-0.009	0.069	0.077	-0.561**	-0.062	-0.499*
	(0.033)	(0.042)	(0.051)	(0.249)	(0.115)	(0.267)
Dereservation(k=+6)	-0.022	0.069	0.091*	-0.340	-0.065	-0.275
	(0.038)	(0.047)	(0.055)	(0.269)	(0.152)	(0.282)
Firm FEs		✓			✓	
Product ×Year FEs		✓			✓	
Firm Controls		✓			✓	
Obs.		9,100			6,476	
R^2		0.91			0.86	

Panel B: Number of Banking Relationships

	Trade Credit Granted			*Log(Primary Product Price Index)*		
	Less Banks	More Banks	Diff (2)-(1)	Less Banks	More Banks	Diff (4)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	-0.063	-0.007	0.055	-0.341*	-0.033	-0.308
	(0.038)	(0.021)	(0.040)	(0.180)	(0.072)	(0.188)
Dereservation(k=+1)	-0.049	0.036*	0.085**	-0.327*	-0.036	-0.291
	(0.038)	(0.022)	(0.038)	(0.173)	(0.081)	(0.204)
Dereservation(k=+2)	-0.033	0.034*	0.067*	-0.282	0.078	-0.360
	(0.039)	(0.021)	(0.038)	(0.187)	(0.128)	(0.230)
Dereservation(k=+3)	-0.030	0.058	0.088	-0.277	-0.009	-0.268
	(0.040)	(0.038)	(0.057)	(0.248)	(0.129)	(0.273)
Dereservation(k=+4)	-0.034	0.021	0.056	-0.341	0.004	-0.345
	(0.043)	(0.027)	(0.047)	(0.232)	(0.143)	(0.278)
Dereservation(k=+5)	-0.054	0.012	0.066	-0.317	0.093	-0.411
	(0.045)	(0.027)	(0.047)	(0.255)	(0.226)	(0.319)
Dereservation(k=+6)	-0.051	0.029	0.080	-0.216	0.147	-0.362
	(0.051)	(0.041)	(0.059)	(0.290)	(0.240)	(0.338)
Firm FEs		✓			✓	
Product ×Year FEs		✓			✓	
Firm Controls		✓			✓	
Obs.		5,018			3,514	
R^2		0.92			0.88	

Table 6: Financing Trade Credit

This table reports the results of regressions that estimate the response of incumbent firms to a threat of entry. I estimate Column (6) of Table 3 for different dependent variables. Column (1) shows the results with (a) Trade Credit Received (scaled by assets), while Columns (2)-(8) shows results with (b) Bank Credit (scaled by assets). In Columns (3)-(8), I report the results of regressions that estimate the response of incumbent firms to a threat of entry based on the firm's access to credit using the size-age index and prior number of banking relationships as a proxy for financial strength/access to finance. I estimate Column (2) of this tables after replacing each *Dereservation(k)* (and *Pre-Dereservation(-k)*) with the interaction terms *Dereservation (k)* $\times$ *Low SA/Less Banks* and *Dereservation (k)* $\times$ *High SA/More Banks*, where *Low SA* is a dummy variable that identifies firms with below-median *Size-Age* Index in the year before the dereservation of its primary product. *Less Banks* identifies the firms with below-median *banking relationships* in the year before the dereservation of their primary product. In this specification, I also include a full set of interaction terms between different groups of financial strength and time fixed effects. This allows two groups of firms to have a differential time trend. Also, I control for industry–specific time fixed effects at the NIC 4-digit level, and I add firm controls for each group separately. In the column titled *Diff*, I test whether the coefficients estimated for different groups are significantly different. Note that in this test, my sample is confined to treated firms that exist one year before the dereservation. Standard errors are corrected for heteroskedasticity and autocorrelation, clustered at the NIC 5-digit product level, and reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1%, respectively.

	Trade Credit Received	*Bank Credit (Credit from Banks and FIs scaled by total assets)*						
			Low SA	High SA	Diff (4)-(3)	Less Banks	More Banks	Diff (7)-(6)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dereservation(k=0)	.005	-.001	0.002	0.086**	0.083**	-0.008	0.023	0.032
	(.007)	(.017)	(0.018)	(0.039)	(0.041)	(0.052)	(0.020)	(0.050)
Dereservation(k=+1)	.009	-.007	-0.029	0.089**	0.118***	-0.002	0.041	0.044
	(.010)	(.019)	(0.018)	(0.037)	(0.038)	(0.049)	(0.030)	(0.056)
Dereservation(k=+2)	.016	-.010	-0.038	0.089**	0.127***	-0.040	0.056**	0.095*
	(.010)	(.025)	(0.037)	(0.038)	(0.045)	(0.052)	(0.024)	(0.051)
Dereservation(k=+3)	.018	-.007	-0.031	0.084**	0.115**	-0.041	0.045	0.086*
	(.013)	(.027)	(0.040)	(0.039)	(0.052)	(0.055)	(0.028)	(0.051)
Dereservation(k=+4)	.029**	-.022	-0.041	0.084**	0.125**	-0.056	0.030	0.086
	(.015)	(.032)	(0.049)	(0.042)	(0.056)	(0.055)	(0.028)	(0.052)
Dereservation(k=+5)	.0008	-.034	-0.065	0.048	0.113*	-0.070	0.006	0.077
	(.017)	(.032)	(0.044)	(0.045)	(0.058)	(0.060)	(0.044)	(0.062)
Dereservation(k=+6)	.007	-.018	-0.023	0.050	0.073	-0.075	0.024	0.099
	(.018)	(.033)	(0.054)	(0.044)	(0.064)	(0.054)	(0.037)	(0.060)
Firm FEs	✓	✓		✓			✓	
Product ×Year FEs	✓	✓		✓			✓	
Firm Controls	✓	✓		✓			✓	
Obs.	9,395	8,970		8,522			4,746	
R^2	0.73	0.65		0.89			0.90	

Table 7: Trade Credit Received and Price of Inputs

This table presents the estimates from difference-in-differences regressions for accounts payable/price of inputs around the removal of entry barriers (dereservation of products). The estimates are based on the regression Equation (1), in which I define *treatment* as the elimination of a small-scale reservation law on the incumbent firm's primary *input* in the NIC 5-digit category. A firm's primary *input* is based on maximum *raw material expenditure* in the year preceding dereservation. *Pre-Dereservation(k)*$_{it}$ (*Dereservation(k)*$_{it}$) is a dummy variable that takes a value one if it is k years before (after) the incumbent firm's primary reserved *input* has been dereserved, and zero otherwise. *Incumbent firm* is defined as a firm that used a reserved product as its primary input before it was dereserved. *Dereservation(k)*$_{it}$ is a dummy that identifies the group of firms with more competition in the input market (or the exposure of firm by using the proportion of total raw material expenditure from dereserved products in the pre-dereservation period). Panel A shows the results with *accounts payable/total assets* as a dependent variable (except Columns (4) & (5)). In Column (1), control group includes only pre-event observations of firms using reserved products that were dereserved later (incumbents). In Column (2), I include firms whose primary inputs were never reserved/dereserved and include an interaction between the year and incumbent dummies (incumbent × year fixed effects). In Column (3), to estimate the exposure of firm to dereservation, I replace dereservation dummies with the proportion of total raw material expenditure from dereserved product in pre-dereservation period. Column (4) is similar to Column (3), except with Log(1+Accounts Payable) as a dependent variable. Column (5) is similar to Column (3), except with *Accounts Payable scaled by Cost of Goods Sold* as a dependent variable. To Column(3), I further add the *one year lagged* value of firm size (measured as log sales), cash holdings, profitability and sales growth as controls and Column (6) shows the results. All regressions are with *incumbent × 4-digit product × year* fixed effects (except Column (1)). Here, I present estimated coefficients for Pre-Dereservation (k=-3 to k=-2) to Dereservation (k=0 to k=+6). Panel B shows results from difference-in-differences regressions for a sample firm's Primary Input Price Index, around the removal of entry barriers (dereservation of products). The specifications are similar to Table 4 with *Log(Primary Input Price Index)* as dependent variable, where *Dereservation(k)*$_{it}$ is defined based on a firm's input rather than output. Panel C of this table, like Table 5, shows the results of regressions that estimate the differential response of incumbent firms to the dereservation of primary input based on their access to finance. For the size-age index, I estimate Column (6) of Panel A after replacing each *Dereservation(k)* (and *Pre-Dereservation(-k)*) with the interaction terms *Dereservation (k) × Low SA* and *Dereservation (k) × High SA*, where *Low SA* is a dummy variable that identifies firms with a below-median *Size-Age* Index in the year before the dereservation of its primary input. Columns (1)-(3) show the results with *Accounts Payable over Total Assets* as a dependent variable, while Columns (4)-(6) show the results with *Log(Primary Input Price Index)* adjusted for inflation, as a dependent variable. Standard errors are corrected for heteroskedasticity and autocorrelation, clustered at NIC 5-digit product level, and reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1%, respectively. All the variables are winsorized at 1%.

Panel A: Trade Credit Received

	Trade Credit Received (*Accounts Payable over Total Assets*)					
	Treated Firms	All Firms	Exposure	Logged	Scaled by COGS	With Controls
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=-3)	0.012	0.008	0.026**	0.163	0.029	0.026**
	(0.009)	(0.009)	(0.011)	(0.101)	(0.019)	(0.011)
Dereservation(k=-2)	0.002	-0.001	0.003	-0.121	-0.007	0.003
	(0.009)	(0.009)	(0.009)	(0.080)	(0.013)	(0.009)
Dereservation(k=0)	0.023**	0.016*	0.019*	0.113	0.019	0.019*
	(0.009)	(0.009)	(0.010)	(0.085)	(0.013)	(0.010)
Dereservation(k=+1)	0.024***	0.016**	0.024**	0.195**	0.029**	0.025**
	(0.008)	(0.007)	(0.010)	(0.078)	(0.012)	(0.010)
Dereservation(k=+2)	0.024***	0.015**	0.034***	0.234***	0.044***	0.032***
	(0.009)	(0.006)	(0.010)	(0.086)	(0.016)	(0.010)
Dereservation(k=+3)	0.034***	0.022**	0.029**	0.331***	0.024	0.027**
	(0.011)	(0.009)	(0.013)	(0.088)	(0.016)	(0.012)
Dereservation(k=+4)	0.023*	0.012	0.035*	0.370***	0.049**	0.033*
	(0.013)	(0.009)	(0.019)	(0.118)	(0.020)	(0.018)
Dereservation(k=+5)	0.006	0.005	0.004	-0.014	0.006	0.004
	(0.014)	(0.011)	(0.016)	(0.121)	(0.015)	(0.016)
Dereservation(k=+6)	0.010	0.004	0.010	0.013	0.010	0.009
	(0.016)	(0.011)	(0.018)	(0.149)	(0.016)	(0.018)
Firm FEs	✓	✓	✓	✓	✓	✓
Incumbent × 4-digit Product×Year FEs	✓	✓	✓	✓	✓	✓
Firm Controls						✓
Obs.	12,022	38,323	38,323	38,323	38,323	38,323
R^2	0.708	0.71	0.71	0.882	0.701	0.712

Panel B: Price of Inputs

	Log(Primary Input Price Index)					
				Adjusted for Inflation		
	Dummies	Exposure	Firm Controls	Dummies	Exposure	Firm Controls
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=-3)	0.064	-0.006	-0.007	0.066	-0.006	-0.007
	(0.055)	(0.067)	(0.068)	(0.056)	(0.067)	(0.068)
Dereservation(k=-2)	0.010	-0.022	-0.022	0.007	-0.027	-0.027
	(0.060)	(0.096)	(0.095)	(0.059)	(0.094)	(0.093)
Dereservation(k=0)	-0.019	-0.059	-0.061	-0.012	-0.046	-0.048
	(0.057)	(0.069)	(0.069)	(0.056)	(0.068)	(0.068)
Dereservation(k=+1)	-0.129**	-0.156**	-0.161**	-0.131**	-0.151**	-0.157**
	(0.060)	(0.078)	(0.078)	(0.056)	(0.075)	(0.076)
Dereservation(k=+2)	-0.114	-0.099	-0.105	-0.111	-0.087	-0.094
	(0.074)	(0.087)	(0.088)	(0.069)	(0.086)	(0.086)
Dereservation(k=+3)	-0.047	-0.041	-0.048	-0.038	-0.021	-0.029
	(0.076)	(0.090)	(0.090)	(0.070)	(0.086)	(0.087)
Dereservation(k=+4)	-0.113	-0.129	-0.137	-0.115	-0.122	-0.131
	(0.087)	(0.113)	(0.112)	(0.081)	(0.109)	(0.109)
Dereservation(k=+5)	-0.175	-0.099	-0.104	-0.173	-0.096	-0.101
	(0.119)	(0.170)	(0.171)	(0.108)	(0.163)	(0.164)
Dereservation(k=+6)	-0.206*	-0.243	-0.247	-0.199*	-0.228	-0.233
	(0.125)	(0.164)	(0.164)	(0.113)	(0.159)	(0.159)
Firm FEs	✓	✓	✓	✓	✓	✓
Incumbent × 4-digit Product×Year FEs	✓	✓	✓	✓	✓	✓
Firm Controls			✓			✓
Obs.	31,458	31,458	31,458	31,458	31,458	31,458
R^2	0.743	0.743	0.743	0.767	0.767	0.767

Panel C: Size–Age Index

	Trade Credit Received			*Log(Primary Input Price Index)*		
	Low SA	High SA	Diff (2)-(1)	Low SA	High SA	Diff (4)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	0.035**	0.018	0.017	0.006	-0.153	-0.159
	(0.018)	(0.017)	(0.031)	(0.108)	(0.247)	(0.276)
Dereservation(k=+1)	0.063**	0.009	0.054	0.026	-0.251	-0.277
	(0.028)	(0.017)	(0.036)	(0.095)	(0.246)	(0.271)
Dereservation(k=+2)	0.032*	0.018	0.015	0.123	-0.214	-0.337
	(0.018)	(0.016)	(0.028)	(0.126)	(0.222)	(0.261)
Dereservation(k=+3)	0.021	0.003	0.018	0.155	-0.278	-0.434*
	(0.020)	(0.020)	(0.032)	(0.115)	(0.210)	(0.251)
Dereservation(k=+4)	-0.009	0.026	-0.036	0.056	-0.328	-0.383
	(0.040)	(0.021)	(0.038)	(0.133)	(0.230)	(0.265)
Dereservation(k=+5)	-0.039	0.006	-0.046	0.235	-0.334	-0.569
	(0.025)	(0.022)	(0.040)	(0.189)	(0.311)	(0.380)
Dereservation(k=+6)	-0.036	0.002	-0.038	-0.051	-0.580**	-0.529*
	(0.028)	(0.024)	(0.044)	(0.168)	(0.247)	(0.317)
Firm FEs		✓			✓	
Product ×Year FEs		✓			✓	
Firm Controls		✓			✓	
Obs.		7,464			4,832	
R^2		0.90			0.81	

Table 8: Priority Sector Lending

This table reports the results of regressions that estimate the differential response of incumbent firms to a threat of entry based on their access to priority sector credit. I re-estimate Column (6) of Table 3 after replacing each *Dereservation(k)* (and *Pre-Dereservation(-k)*) with interaction terms *Dereservation (k)* × *PSL* and *Dereservation (k)* × *Non-PSL,* where *PSL* is a dummy variable that identifies firms in priority sector in the year before the dereservation of their primary product. In this specification, I also include a full set of interaction terms between different groups of access to finance and time fixed effects. This allows two groups of firms to have a differential time trend. Also, I control for industry–specific time fixed effects at NIC 4-digit level, and I add firm controls, for each group separately. In the column titled *Diff*, I test whether the coefficients estimated for different groups are significantly different. Note that in this test, my sample is confined to treated firms that exist one year before the dereservation. Columns (1)-(3) show the results with *Accounts Receivables over Sales* as a dependent variable, while Columns (4)-(6) show the results with *Log(Primary Product Price Index)*, adjusted for inflation, as a dependent variable. Standard errors are corrected for heteroskedasticity and autocorrelation, clustered at NIC 5-digit product level, and reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1%, respectively.

Priority Sector Lending

	Trade Credit Granted			*Log(Primary Product Price Index)*		
	Non-Priority Sector	Priority Sector	Diff (2)-(1)	Non-Priority Sector	Priority Sector	Diff (5)-(4)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	0.005	-0.021	-0.027	-0.079	-0.369***	-0.291**
	(0.012)	(0.062)	(0.058)	(0.054)	(0.125)	(0.143)
Dereservation(k=+1)	0.019	0.095**	0.076*	-0.120**	-0.221	-0.101
	(0.012)	(0.040)	(0.040)	(0.051)	(0.162)	(0.173)
Dereservation(k=+2)	0.023*	0.137*	0.113	-0.107	-0.169	-0.062
	(0.012)	(0.082)	(0.084)	(0.070)	(0.221)	(0.236)
Dereservation(k=+3)	0.029*	0.031	0.002	-0.090	-0.267	-0.177
	(0.018)	(0.079)	(0.082)	(0.105)	(0.197)	(0.216)
Dereservation(k=+4)	0.021	0.040	0.019	-0.083	-0.390*	-0.307
	(0.017)	(0.079)	(0.081)	(0.089)	(0.199)	(0.210)
Dereservation(k=+5)	0.041*	-0.091	-0.132	-0.155	-0.877**	-0.722
	(0.022)	(0.087)	(0.092)	(0.105)	(0.409)	(0.439)
Dereservation(k=+6)	0.026	-0.074	-0.100	-0.054	-0.698**	-0.644*
	(0.025)	(0.081)	(0.085)	(0.159)	(0.292)	(0.355)
Firm FEs		✓			✓	
Product ×Year FEs		✓			✓	
Firm Controls		✓			✓	
Obs.		8,075			5,797	
R^2		0.91			0.86	

Table 9: Entry Deterrence

This table reports the results of regressions that estimate the differential effect of trade credit on entry rates. Here, I estimate an entry rate as the ratio of the number of entrants to the number of incumbents in a 5-digit industry for a given year. I then estimate the baseline specification for the group of firms that have increased (decreased) trade credit in the year of dereservationand and one year after the dereservation. The regression also include 5-digit industry fixed effects and year fixed effects. Standard errors are corrected for heteroskedasticity and autocorrelation, clustered at NIC 5-digit product level, and reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1%, respectively.

	Entry Rate		
	Δ Trade Credit Granted > 0 k=0,1	Δ Trade Credit Granted ≤ 0 k=0,1	Diff
	(1)	(2)	(1)-(2)
Dereservation(k=0)	.058	.148***	-.09
	(.039)	(.057)	(.065)
Dereservation(k=+1)	.212***	.407***	-.196**
	(.055)	(.092)	(.093)
Dereservation(k=+2)	.315***	.533***	-.218*
	(.092)	(.093)	(.116)
Dereservation(k=+3)	.402***	.646***	-.244***
	(.095)	(.110)	(.093)
Dereservation(k=+4)	0.850***	0.873***	-0.023
	(0.160)	(0.168)	(0.136)
Dereservation(k=+5)	0.887***	1.087***	-0.200
	(0.185)	(0.207)	(0.153)
Dereservation(k=+6)	1.049***	1.281***	-0.232
	(0.211)	(0.269)	(0.146)
Industry Fixed Effects	5-digit		
Year Fixed Effects	✓		
Obs.	1,900		
R^2	0.57		

Table 10: Robustness Checks

This table provides the robustness of the results for the baseline specification (i.e., Column (6) of Table 3). Panel A shows the results for *Accounts Receivables over Sales* as a dependent variable, while Panel B shows the results for *Log(Primary Product Price Index)*. Major changes in policies vis-a-vis foreign investment occurred in the early 1990s and then stalled during the period of dereservation reform (i.e., after 1997). Column (1) show the results for the sample beginning in 1997. Most of the products in sample dereserved after 2003. Column (2) reports the results for the sample beginning in 2003. In Column (3), I exclude firms that manufacture chemical products. Column (4) includes state-year fixed effects. In Column (5), I redefine the incumbent as a firm that manufactured a reserved product as its primary product three years before it was dereserved. In Column (6), I exclude industries (at the 5-digit) with large leader firms. Standard errors are corrected for heteroskedasticity and autocorrelation, clustered at NIC 5-digit product level, and reported in parentheses. Finally, in Column (7), I correct standard errors for clusters at firm-level. *,**, and *** indicate significance at 10%, 5%, and 1%, respectively.

Panel A: Trade Credit Granted

	Accounts Receivables over Sales						
	After 1997	After 2003	Exclude chemical products	State-year FEs	Treatment definition	Exclude Industries with Large Firms	Standard Errors
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dereservation(k=-3)	0.002	0.001	0.00003	0.006	0.002	-0.0006	0.004
	(0.012)	(0.014)	(0.012)	(0.011)	(0.012)	(0.020)	(0.012)
Dereservation(k=-2)	0.012	0.019	0.015	0.014	0.013	-0.009	0.016
	(0.012)	(0.012)	(0.015)	(0.014)	(0.014)	(0.028)	(0.012)
Dereservation(k=0)	0.012	0.024**	0.013	0.014	0.013	0.011	0.015
	(0.011)	(0.009)	(0.012)	(0.011)	(0.012)	(0.020)	(0.011)
Dereservation(k=+1)	0.023**	0.036***	0.023*	0.027**	0.020	0.043*	0.026**
	(0.012)	(0.012)	(0.012)	(0.013)	(0.012)	(0.024)	(0.012)
Dereservation(k=+2)	0.019*	0.033***	0.018	0.023**	0.016	0.037**	0.022*
	(0.011)	(0.012)	(0.013)	(0.011)	(0.011)	(0.017)	(0.013)
Dereservation(k=+3)	0.028*	0.047***	0.024*	0.035**	0.026**	0.057**	0.034**
	(0.015)	(0.016)	(0.014)	(0.015)	(0.013)	(0.029)	(0.015)
Dereservation(k=+4)	0.026*	0.043**	0.023	0.030**	0.014	0.049*	0.031**
	(0.015)	(0.018)	(0.016)	(0.015)	(0.015)	(0.029)	(0.015)
Dereservation(k=+5)	0.029*	0.051***	0.031*	0.035*	0.018	0.064**	0.034*
	(0.017)	(0.019)	(0.018)	(0.018)	(0.016)	(0.028)	(0.018)
Dereservation(k=+6)	0.023	0.048**	0.019	0.028	0.011	0.056	0.029
	(0.020)	(0.022)	(0.018)	(0.020)	(0.018)	(0.039)	(0.019)
Firm FEs	✓	✓	✓	✓	✓	✓	✓
Incumbent × 4-digit Product×Year FEs	✓	✓	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓	✓	✓
Obs.	26,018	18,294	26,728	31,008	31,161	24,594	31,161
R^2	0.727	0.772	0.714	0.723	0.709	0.715	0.709

Panel B: Price of Output

	Log(Primary Product Price Index)						
	After 1997	After 2003	Exclude chemical products	State-year FEs	Treatment definition	Exclude Industries with Large Firms	Standard Errors
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dereservation(k=-3)	-0.041	-0.061	-0.071	-0.053	-0.038	-0.083	-0.051
	(0.043)	(0.060)	(0.054)	(0.045)	(0.042)	(0.062)	(0.051)
Dereservation(k=-2)	-0.019	0.020	-0.071	-0.019	-0.018	-0.047	-0.032
	(0.041)	(0.045)	(0.056)	(0.045)	(0.046)	(0.062)	(0.047)
Dereservation(k=0)	-0.071	-0.051	-0.143***	-0.091*	-0.071	-0.112	-0.086
	(0.048)	(0.055)	(0.052)	(0.055)	(0.051)	(0.087)	(0.057)
Dereservation(k=+1)	-0.111***	-0.106*	-0.128***	-0.115**	-0.078*	-0.194***	-0.121**
	(0.040)	(0.054)	(0.049)	(0.046)	(0.043)	(0.070)	(0.056)
Dereservation(k=+2)	-0.087	-0.081	-0.153**	-0.095	-0.051	-0.137*	-0.105*
	(0.058)	(0.078)	(0.075)	(0.065)	(0.063)	(0.074)	(0.061)
Dereservation(k=+3)	-0.057	-0.058	-0.090	-0.059	-0.022	-0.177**	-0.082
	(0.080)	(0.104)	(0.107)	(0.085)	(0.076)	(0.083)	(0.064)
Dereservation(k=+4)	-0.085	-0.091	-0.094	-0.059	-0.038	-0.227**	-0.099
	(0.071)	(0.103)	(0.081)	(0.082)	(0.074)	(0.112)	(0.068)
Dereservation(k=+5)	-0.180**	-0.170	-0.174*	-0.153	-0.165*	-0.373***	-0.206**
	(0.088)	(0.115)	(0.104)	(0.094)	(0.084)	(0.122)	(0.087)
Dereservation(k=+6)	-0.073	-0.097	0.021	-0.023	-0.023	-0.411***	-0.093
	(0.117)	(0.146)	(0.108)	(0.127)	(0.101)	(0.154)	(0.095)
Firm FEs	✓	✓	✓	✓	✓	✓	✓
Incumbent × 4-digit Product×Year FEs	✓	✓	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓	✓	✓
Obs.	19,045	12,672	19,890	23,324	23,500	18,956	23,500
R^2	0.787	0.826	0.788	0.799	0.788	0.799	0.788

Appendix A

Definitions

Variables	Definition
Trade Credit Granted	Account receivables (*trade_receivables*) divided by total sales(*sales*)
Price of Output	For each product manufactured by sample firms, I create a price index and use the *logged* value of main product's price index.
Trade Credit Received	Account payable (*sundry_creditors*) divided by total assets(*total_assets*)
Price of Input	For each input used by the sample firms, I create a price index and use the *logged* value of main product's price index.
Cash Holdings	Cash (*cash*) divided by total assets(*total_assets*)
Profitability	EBITDA (*pbitda*) divided by total assets(*total_assets*)
Sales Growth	Sales-Lagged Sales divided by Lagged Sales
Bank Credit	Borrowing from banks and financial institutions *(bank_borrowings+ fin_inst_borr+ borr_syndicated_banks_inst)* scaled by total assets

Internet Appendix

Financial Constraints and Trade Credit as a Strategic Tool: Evidence from Small-Scale Reservation Reforms in India

Manpreet Singh

IA1 Matching Product Data

The reporting of products by Indian firms is not governed by any particular product classification. However, CMIE has developed an internal product classification that is based on the Harmonized System (HS) and National Industry Classification (NIC) schedules. There are about 3,514 codes based on the NIC and the HS schedule. CMIE has explicitly linked the product names reported by the firms to this classification. The names of products reported by the firm could differ in aggregation, or even in spelling (e.g., "Steel Rod" versus "Steel Rods"). Most of these issues were addressed by CMIE. But, some discrepancies remain, such as like "Millstone, grindstone" versus "Millstone, grindstone, etc.", which I reconciled manually. CMIE has standardized approximately 51,320 product names to 3,714 possible CMIE product codes.

My final sample includes 2,710 product codes out of the universe of 3,714 product codes. The unused codes are products in the agriculture and services sectors, which, of course, are not produced by manufacturing firms. These 2,710 product codes are mapped to 591 5-digit NIC product codes in 137 4-digit NIC industries across the 24 manufacturing sectors (2-digit NIC codes). As a comparison, the US data used by Bernard, Redding and Schott (2011) contain approximately 1,500 products defined as 5-digit Standard Industrial Classification (SIC) codes, across 455 4-digit SIC industries. Thus, my definition of a product is slightly more detailed than that of Bernard, Redding and Schott (2011).

I created a concordance between the NIC product codes and the list of reserved and dereserved products. I matched products reserved to each firm based on the 5-digit industry. In most cases, the match between the NIC codes and SSI codes/product codes was so exact that I could match them based solely on the product descriptions. In other cases, I used the lengthy descriptions associated with the industry codes to help resolve many questionable concordances. I assumed that a product was matched to an NIC code if it was at least a partial match. Table IA1 shows a sample notification about a list of items dereserved in 1997, while Table IA2 provides information on the number of products dereserved in a 2-digit NIC industry for the period 2003-2008. Next, Table IA3 shows the mapping of product codes in a dereserved list with NIC codes. All the notifications about dereservation are available at http://www.dcmsme.gov.in/publications/notiissforresderes.html.

Examples of products within the Food Product sector (NIC 10) of this hierarchical mapping are listed in Table IA4. The table lists two industries in the sector: Manufacture of Bakery Products, which contains three products (at the NIC 5-digit level), and Manufacturing of Sugar, which contains nine products. The product classification provides a concordance with the more familiar NIC industry codes used to classify economic activity in India. Each of the 2,710 product codes can therefore be mapped to a 5-, 4-, 2-, 2-, or 1-digit NIC code. Several features of these product data give me additional confidence in their quality despite the self-reported and nonstandardized nature of the data set. First, firms are required to report not just the names of the products, but also product-level details about installed capacity, production, sales quantity, and value. The product-level data are available for 85 % of the firms; this accounts for more than 90% of the output and exports of the manufacturing firms in Prowess. More importantly, the product-level information and overall output are in separate modules of the Prowess database, and this enables me to cross–check the consistency of the data. The data span the period from 1990 to 2013.

IA2 Sample Selection

For my sample period, I start with a sample of unique 24,165 firms with nonmissing assets and sales. Given that the dereservation rule applies to only manufacturing firms, I omit all the non-manufacturing firms: I keep only those firms that have an NIC 2-digit code between 10 and 33. This reduced the numbers of sample firms to 9,937. Later, I excluded all firms owned by the state and central government (about 270 firms). Out of 9,667, product data were available for almost two-thirds of the firms (6,771 firms). Of these frims, I was able to correctly match the dereservation product codes for 4,600 firms. The treated sample (i.e., firms that manufacture a reserved product as its main product) consists of almost 1,654 firms. Regression results based on this treated sample are reported in Column 1 of Table 3, which accounts for almost 13,989 firm-year observations. As mentioned before, Indian firms are required by the 1956 Companies Act to disclose product-level information on capacities, production, and sales in their annual reports. To estimate the price index, I used information on sales quantity and divide the total sales by the number of units of the product that were sold. But, due to inconsistencies in the data, I could complete the matching for only 4,091 firms. The information on raw materials that can be matched to dereservation product codes was available for almost 6,847 firms. Table A5 provides the definition for the variables used.

IA3 Access to Finance

IA3.1 Debt Recovery Tribunals

In this subsection, I use the establishment of debt recovery tribunals (DRTs) in India as a proxy for access to finance. DRTs reduced the cost of the legal enforcement of debt contracts in different states with no accompanying change in the underlying law. Various studies show that lower enforcement costs are positively associated with the size of new loans and debt maturity, and they are negatively related to interest costs (Lilienfeld-Toal, Mookherjee, and Visaria, 2012; Gopalan, Mukherjee, and Singh, 2016; Visaria, 2009). As in Panel A, Table 5, I re-estimate Column (6) of Table 3 with *Late DRT* and *Early DRT* as interaction terms, where *Early DRT* identifies the state of a firm's registered office where DRTs were operational for at least five years in the year before the dereservation of its primary product. Table IA5, Panel A shows the results. I find that firms located in early DRT states were quick to respond to an entry threat by increasing the credit offered to customers, while incumbent firms located in late DRT states relied on lower prices. I find that firms located in early DRT states increase trade credit by 18.9% (or an increase in the term of credit by 17 days above the previous level of 73 days) and I find that these firms cut prices by 9.2% one year after the dereservation of its main product compared to one year before the dereservation. Firms in late DRT states do not change trade credit terms, and they cut prices by almost 40%.

IA3.2 Relationship with Large Banks

As in Panel B, Table 5, I re-estimate Column (6) of Table 3 with *Small Banks* and *Large Banks* as interaction terms: The variable *Large Banks* identifies firms that have banking relationships with large banks. These banks account for 50% of loans outstanding and the number of relationships in the year before the dereservation of its primary product. Table IA5, Panel B shows the results.[29] I find that firms that have banking relationships with large banks at the time of the entry threat offered more credit to customers.

IA3.3 Access to Bank Credit

A firm's geographic location plays an important roles in its in access to the local financial market. In this subsection, I re-estimate column (6) of Table 3 with *Low Bank Credit* and *High Bank Credit* as interaction terms: The variable *High Bank Credit* identifies firms with a registered office located in a state with above-median per capita bank credit extended one year before the dereservation of their primary product. Table IA5, Panel C shows the results. I find consistent results here as well: Firms with better access to financial markets use trade credit as a strategic tool, while other firms rely on limit-pricing.

IA3.4 Access to Supplier's Credit

Giannetti, Burkart, and Ellingsen (2011) find that firms that borrow more using trade credit also lend more trade credit. I test this prediction in this subsection. I re-estimate Column (6) of Table 3 with *Low Supplier Credit* and *High Supplier Credit* as interaction terms: The variable *Low Supplier Credit* identifies firms with below-median trade payable (scaled by assets) in the year before the dereservation of its primary product. Table IA5, Panel D shows the results. I find consistent results: Firms with better access to credit from suppliers use trade credit as strategic tool while other firms rely on limit-pricing.

[29]Large banks include the State Bank of India, ICICI Bank, Canara Bank, Bank of Baroda, HDFC Bank, Axis Bank, Bank of India, Union Bank of India, the Central Bank of India, Indian Overseas Bank, and the United Bank of India.

IA3.5 Customer's access to credit

As in Table 7, Panel C, I test how an incumbent firm's response to a threat of entry varies with customers, financial strength/access to finance. In Table IA5, Panel E, I report results with DRT as a proxy for financial constraints. I find that firms located in states with late DRT receive more trade credit.

The overall findings in this subsection suggest that financial constraints do impact the choice of strategic tool when firms face an entry threat from new competitors. Firms with deeper pockets offer more credit, and firms with financial constraints take more credit when bargaining power shifts from the supplier to the customer. However, unconstrained customer firms demand lower prices when their suppliers face more competition.

Table IA1: Sample Dereservation Notification

Here I provide a sample notification about a list of items dereserved in 1997. All the notifications about the dereservation of products are available at http://www.dcmsme.gov.in/publications/notiissforresderes.html

रजिस्ट्री सं॰ डी॰ एल॰–33004/97 REGD. NO. D. L.-33004/97

भारत का राजपत्र
The Gazette of India

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MINISTRY OF INDUSTRY
(Department of Industrial Policy and Promotion)

NOTIFICATION

New Delhi, the 3rd April, 1997

S. O. 298 (E).—In exercise of the powers conferred by sub-section (2A) of section 29B of the Industries (Development and Regulation) Act, 1951 (65 of 1951) and after considering the recommendations made by the Advisory Committee constituted under sub-section (2B) of the said section, the Central Government hereby direct that the following amendments shall be made in the notification of the Government of India in the Ministry of Industry, Department of Industrial Development No. S.O.477(E), dated the 25th July, 1991, namely :—

In the said notification, in Schedule-III under the heading—LIST OF ITEMS RESERVED FOR EXCLUSIVE MANUFACTURE IN SMALL SCLAE SECTOR, the following items and the entries relating thereto shall be omitted, namely :—

Sl. No. (As per gazette Notification)	Product Code	Name of the product	
1	2	3	
"1	201801	Ice Cream	
4	202901	Vinegar	
5	204200	Rice Milling	
6	204300	Dal Milling	
8	20530101	Biscuits	
15	21620101	Poultry feed except in pellet form	
16C	224302	Synthetic syrups	
54	280906	Corrugated paper & boards	
571	363708	Hair driers—all types	
629	374717	Hub caps—auto	
635	374737	Ornamental fittings—auto	
644	37478101	Spot lamps assembly—auto	Excluding combination lamp assembly
645	37478201	Stop lamps assembly—auto	
646	3747870	Tail lamp assembly—auto	
647	3747804	Ash trays—Car fittings"	

Table IA2: Timing of Dereservation Reform by Industry-Year

The table below provides information on the number of products dereserved in a 2-digit NIC industry for the period 2003-2008.

		Number of Products Dereserved					
2-digit	**Industry/ Year**	2003	2004	2005	2006	2007	2008
10	Food Products	0	2	0	2	0	6
11	Beverages	0	0	0	0	0	0
12	Tobacco	0	0	0	0	0	0
13	Textiles	0	0	33	0	0	0
14	Wearing Apparel	2	0	0	0	1	0
15	Leather Products	7	0	0	0	0	0
16	Wood Products	0	5	0	0	8	0
17	Paper Products	3	8	0	0	0	17
18	Printing	0	0	0	0	0	1
19	Coke and refined petroleum products	0	0	2	1	0	1
20	Chemicals	57	93	0	50	2	41
21	Pharmaceuticals	0	0	0	7	1	1
22	Rubber and plastics	0	1	20	0	26	11
23	Nonmetallic mineral products	0	3	0	3	39	2
24	Basic Metals	0	0	0	0	0	0
25	Fabricated metal products	0	24	0	45	33	5
26	Computer, electronic, & optical products.	0	2	0	1	6	2
27	Electrical equipment	0	3	28	2	7	11
28	Machinery and equipment	0	5	3	10	13	2
29	Motor vehicles, trailers, & semi-trailers	0	4	0	9	4	0
30	Other transport equipment	0	0	4	35	42	0
31	Furniture	0	1	0	6	2	2
32	Other manufacturing	1	2	1	1	33	4

Table IA3: Sample of Product Matches

This table provides the sample of matches of product codes in the dereserved list with NIC codes and the dereservation dates.

Item No.	Product Code	Product Name	NIC Activity Name	NIC5 Digit	Dereservation Date
1	201801	Ice Cream	Manufacture of ice-cream and kulfi etc.	10505	3 Apr, 1997
3	202501	Pickles & Chutneys	Manufacture of pickles, chutneys, murabbas etc.	10306	10 Apr, 2015
5	204200	Rice Milling	Rice milling	10612	3 Apr, 1997
6	204300	Dal Milling	Dal milling	10613	3 Apr, 1997
7	205101	Bread	Bread making	10711	10 Apr, 2015
242	312405	Chlorinated paraffin wax	Paraffin wax	19202	10 Oct, 2008

Table IA4: Examples of Industries, Sectors, and Products

NIC 1071 contains three products and NIC 1072 contains nine products.

Group	Class	Sub-class	Description
	1071		**Manufacture of bakery products**
			This class excludes:
			- manufacture of farinaceous products (pastas), see 1074
			- manufacture of potato snacks, see 1030
			- heating up of bakery items for immediate consumption, see division 56
		10711	Manufacture of bread
		10712	Manufacture of biscuits, cakes, pastries, rusks etc.
		10719	Manufacture of other bakery products n.e.c.
	1072		**Manufacture of sugar**
			This class excludes:
			- manufacture of glucose, glucose syrup, maltose, see 1062
		10721	Manufacture or refining of sugar (sucrose) from sugercane
		10722	Manufacture of `gur' from sugarcane
		10723	Manufacture of `gur' from other than sugarcane
		10724	Manufacture of `khandsari' sugar from sugarcane
		10725	Manufacture of `khandsari' suger from other than sugarcane
		10726	Manufacture of `boora' and candy from sugarcane
		10727	Manufacture of `boora' and candy from other than sugarcane
		10728	Manufacture of molasses
		10729	Manufacture of sugar from other sources (juice of palm, suger beet etc.)

Table IA5: Access to Finance

This table shows the results of regressions that estimate the differential response of incumbent firms to threats of entry based on their financial strength/access to finance. I use different proxies for access to finance. I estimate Column (6) of Table 3 after replacing each *Dereservation(k)* (and *Pre-Dereservation(-k)*) with interaction terms *Dereservation (k)* × *More Access* and *Dereservation (k)* × *Less Access*. In Panel A, I report results in which I use quasi-natural experiment: The establishment of Debt Recovery Tribunals (DRTs) in India (Gopalan, Mukherjee, and Singh, 2016) as a measure of access to finance. *Early DRT* identifies the state of a firm's registered office where DRTs were operational for at least five years in the year before the dereservation of its primary product. In Panel B, I use *Small Banks* and *Large Banks* as interaction terms: *Large Banks* identifies firms having banking relationships with large banks in the year before the dereservation of its primary product. In Panel C, I use *Low Bank Credit* and *High Bank Credit* as interaction terms: *High Bank Credit* identifies firms with a registered office located in a state with above-median per capita bank credit extended one year before the dereservation of their primary product. In Panel D, I use *Low Supplier Credit* and *High Supplier Credit* as interaction terms: *Low Supplier Credit* identifies firms with below-median trade payable (scaled by assets) in the year before the dereservation of its primary product. In Panel E, I test how incumbent firm's response to a threat of entry varies with customers financial strength/access to finance. I report results with DRT as proxy for financial constraints. In these specifications, I also include a full set of interaction terms between different groups of access to finance and time fixed effects. This allows two groups of firms to have a differential time trend. Also, I control for industry-specific time fixed effects at the NIC 4-digit level, and I add firm controls for each group separately. In the column titled *Diff*, I test whether the coefficients estimated for different groups are significantly different. Note that in this test my sample is confined to treated firms that exist one year before the dereservation. Columns (1)-(3) show the results with *Accounts Receivables over Sales* (except Panel E) as a dependent variable, while Columns (4)-(6) show the results with *Log(Primary Product Price Index)* (except Panel E), adjusted for inflation, as a dependent variable. Standard errors are corrected for herosKedasticity and autocorrelation, clustered at the NIC 5-digit product level, and reported in parentheses. *,**, and *** indicate significance at 10%, 5%, and 1%, respectively.

Panel A: Establishment of Debt Recovery Tribunals (DRTs)

	Trade Credit Granted			*Log(Primary Product Price Index)*		
	Late DRT	Early DRT	Diff (2)-(1)	Late DRT	Early DRT	Diff (4)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	-0.001	0.036	0.037	-0.435*	-0.091	-0.345
	(0.017)	(0.028)	(0.029)	(0.240)	(0.060)	(0.241)
Dereservation(k=+1)	0.016	0.049*	0.033	-0.508**	-0.114**	-0.394
	(0.018)	(0.028)	(0.029)	(0.241)	(0.046)	(0.247)
Dereservation(k=+2)	0.028	0.050*	0.022	-0.484*	-0.109	-0.375
	(0.020)	(0.029)	(0.029)	(0.250)	(0.074)	(0.259)
Dereservation(k=+3)	0.050*	0.045	-0.006*	-0.574**	-0.081	-0.493**
	(0.026)	(0.031)	(0.032)	(0.269)	(0.112)	(0.245)
Dereservation(k=+4)	0.059**	0.035	-0.023	-0.579**	-0.074	-0.505*
	(0.025)	(0.033)	(0.032)	(0.258)	(0.101)	(0.260)
Dereservation(k=+5)	0.051*	0.035	-0.016	-0.694***	-0.135	-0.558**
	(0.028)	(0.035)	(0.034)	(0.250)	(0.102)	(0.252)
Dereservation(k=+6)	0.055	0.029	-0.026	-0.669**	0.019	-0.688**
	(0.035)	(0.037)	(0.038)	(0.285)	(0.123)	(0.294)
Firm FEs		✓			✓	
Product ×Year FEs		✓			✓	
Firm Controls		✓			✓	
Obs.		9,448			6,793	
R^2		0.91			0.84	

Panel B: Relationship With Large Banks

	Trade Credit Granted			*Log(Primary Product Price Index)*		
	Small Banks	Large Banks	Diff (2)-(1)	Small Banks	Large Banks	Diff (4)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	-0.044	0.007	0.051*	-0.121	-0.106	-0.014
	(0.029)	(0.016)	(0.030)	(0.124)	(0.191)	(0.232)
Dereservation(k=+1)	0.000	0.031	0.031	-0.118	-0.157	0.040
	(0.037)	(0.019)	(0.047)	(0.153)	(0.177)	(0.229)
Dereservation(k=+2)	0.007	0.040**	0.033	-0.132	-0.040	-0.093
	(0.030)	(0.019)	(0.038)	(0.270)	(0.180)	(0.351)
Dereservation(k=+3)	0.025	0.050*	0.025	-0.332	-0.089	-0.243
	(0.058)	(0.029)	(0.063)	(0.356)	(0.196)	(0.422)
Dereservation(k=+4)	0.042	0.027	-0.015	-0.409	-0.086	-0.323
	(0.047)	(0.024)	(0.049)	(0.423)	(0.218)	(0.478)
Dereservation(k=+5)	0.037	0.000	-0.037	-0.300	-0.123	-0.177
	(0.054)	(0.029)	(0.055)	(0.420)	(0.217)	(0.480)
Dereservation(k=+6)	0.045	0.014	-0.031	-0.187	-0.152	-0.035
	(0.059)	(0.037)	(0.059)	(0.489)	(0.225)	(0.554)
Firm FEs		✓			✓	
Product ×Year FEs		✓			✓	
Firm Controls		✓			✓	
Obs.		5,117			4,357	
R^2		0.92			0.85	

Panel C: Access to Bank Credit

	Trade Credit Granted			*Log(Primary Product Price Index)*		
	Low Bank Credit	High Bank Credit	Diff (2)-(1)	Low Bank Credit	High Bank Credit	Diff (4)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	-0.001	0.006	0.007	-0.428**	-0.042	-0.387**
	(0.039)	(0.015)	(0.034)	(0.169)	(0.080)	(0.188)
Dereservation(k=+1)	0.004	0.024	0.021	-0.470***	-0.066	-0.404**
	(0.040)	(0.016)	(0.038)	(0.174)	(0.078)	(0.182)
Dereservation(k=+2)	-0.006	0.025	0.032	-0.459***	0.005	-0.464**
	(0.040)	(0.017)	(0.039)	(0.167)	(0.113)	(0.211)
Dereservation(k=+3)	0.005	0.038*	0.034	-0.376**	0.033	-0.409
	(0.041)	(0.020)	(0.036)	(0.189)	(0.147)	(0.253)
Dereservation(k=+4)	-0.012	0.043**	0.055	-0.448**	-0.009	-0.439*
	(0.039)	(0.021)	(0.034)	(0.190)	(0.158)	(0.248)
Dereservation(k=+5)	0.006	0.005	-0.000	-0.457**	-0.172	-0.285
	(0.040)	(0.026)	(0.036)	(0.226)	(0.169)	(0.303)
Dereservation(k=+6)	0.002	0.015	0.013	-0.447**	-0.106	-0.341
	(0.043)	(0.030)	(0.038)	(0.223)	(0.205)	(0.313)
Firm FEs		✓			✓	
Product ×Year FEs		✓			✓	
Firm Controls		✓			✓	
Obs.		8,841			7,352	
R^2		0.91			0.84	

Panel D: Access to Supplier's Credit

	Trade Credit Granted			*Log(Primary Product Price Index)*		
	Low Supplier Credit	High Supplier Credit	Diff (2)-(1)	Low Supplier Credit	High Supplier Credit	Diff (4)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	0.011	0.019	0.008	-0.330**	-0.185*	-0.145
	(0.027)	(0.020)	(0.033)	(0.151)	(0.097)	(0.154)
Dereservation(k=+1)	0.018	0.036*	0.018	-0.316*	-0.236**	-0.080
	(0.023)	(0.020)	(0.030)	(0.160)	(0.092)	(0.174)
Dereservation(k=+2)	0.021	0.049**	0.028	-0.361*	-0.195**	-0.166
	(0.028)	(0.023)	(0.036)	(0.206)	(0.085)	(0.230)
Dereservation(k=+3)	0.011	0.069**	0.058	-0.410*	-0.069	-0.341
	(0.029)	(0.032)	(0.043)	(0.242)	(0.132)	(0.260)
Dereservation(k=+4)	-0.014	0.078**	0.092**	-0.392	-0.128	-0.263
	(0.030)	(0.030)	(0.038)	(0.252)	(0.122)	(0.283)
Dereservation(k=+5)	-0.017	0.057	0.074*	-0.480*	-0.260**	-0.220
	(0.031)	(0.036)	(0.044)	(0.250)	(0.124)	(0.280)
Dereservation(k=+6)	-0.039	0.075*	0.115**	-0.423	-0.114	-0.309
	(0.034)	(0.045)	(0.053)	(0.262)	(0.229)	(0.355)
Firm FEs		✓			✓	
Product ×Year FEs		✓			✓	
Firm Controls		✓			✓	
Obs.		7,352			6,366	
R^2		0.84			0.85	

Panel E: Customer's Financial Constraint, Establishment of Debt Recovery Tribunals (DRTs)

	Trade Credit Received			*Log(Primary Input Price Index)*		
	Late DRT	Early DRT	Diff (2)-(1)	Late DRT	Early DRT	Diff (4)-(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Dereservation(k=0)	0.132***	0.018	0.114***	0.012	-0.057	0.069
	(0.038)	(0.011)	(0.038)	(0.280)	(0.079)	(0.296)
Dereservation(k=+1)	0.119***	0.035**	0.084**	0.128	-0.089	0.217
	(0.035)	(0.014)	(0.037)	(0.269)	(0.079)	(0.277)
Dereservation(k=+2)	0.116***	0.025*	0.091**	0.131	-0.015	0.147
	(0.040)	(0.014)	(0.041)	(0.302)	(0.083)	(0.324)
Dereservation(k=+3)	0.080*	0.008	0.072	0.173	-0.048	0.220
	(0.044)	(0.015)	(0.045)	(0.309)	(0.097)	(0.339)
Dereservation(k=+4)	0.085*	0.016	0.068	0.207	-0.104	0.311
	(0.045)	(0.023)	(0.046)	(0.363)	(0.099)	(0.382)
Dereservation(k=+5)	0.012	0.011	0.001	0.166	-0.017	0.182
	(0.041)	(0.015)	(0.044)	(0.442)	(0.147)	(0.485)
Dereservation(k=+6)	-0.001	0.015	-0.016	0.190	-0.346*	0.537
	(0.043)	(0.019)	(0.047)	(0.473)	(0.194)	(0.494)
Firm FEs		✓			✓	
Product ×Year FEs		✓			✓	
Firm Controls		✓			✓	
Obs.		6,757			4,757	
R^2		0.90			0.81	